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Hybrid Fuzzy Machine Learning Models Optimized with Meta-Heuristics for Accurate EEG-Based Neurological Assessment

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Abstract
Accurate and timely analysis of electroencephalogram (EEG) signals is critical for the assessment of neurological disorders such as coma and epileptic seizures. Conventional EEG analysis is often time-consuming, prone to human error, and limited by the availability of skilled specialists, highlighting the need for automated, reliable, and intelligent diagnostic systems. This study presents a unified hybrid framework that leverages meta-heuristic optimized machine learning approaches for the classification of EEG signals in multiple neurological conditions. Features were extracted from EEG signals, including time- and frequency-domain characteristics, statistical properties, and nonlinear metrics. Feature mapping and dimensionality reduction were performed using advanced optimization techniques such as Harris Hawks Optimization (HHO) and the Starfish Optimization Algorithm (SFOA), combined with Fuzzy-PCA and Auto-Encryption (AE) for robust feature representation. Classification was conducted using hybrid models including Fuzzy K-NN, FSVM, and DT-FIS, enabling accurate discrimination between different levels of consciousness and stages of epileptic seizures. Experimental results demonstrated high performance, achieving up to 99.53% accuracy for deep coma classification and 99.28% F1-score for seizure detection, with significant improvements in precision, recall, and robustness against feature variability. The proposed framework highlights the efficacy of combining hybrid learning models, fuzzy logic, and meta-heuristic optimization for EEG-based diagnosis, providing a scalable, automated, and highly accurate system for neurological assessment.

Keywords: Epilepsy Diagnosis, EEG Signals, Feature Reduction Matrix, DT-FIS Machine Learning System, Starfish Optimization Algorithm (SFOA), Water Cycle Algorithm-automatic Encoder (WCA-AE), Harris Hawks Optimization (HHO).

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1. Introduction

EEG was first recorded by Hans Berger in 1924, revolutionizing neurology by enabling non-invasive brain activity monitoring. Its clinical applications have evolved from diagnosing epilepsy and sleep disorders to real-time seizure detection and coma assessment, forming the basis for modern CAD systems like ours. Over 70 million individuals worldwide suffer from epilepsy, a chronic, non-communicable brain illness, according to the World Health Organization (WHO). Timely identification and management of epileptic seizures, a major cause of death and morbidity globally, present considerable obstacles [1]. Throughout a person's life, they may experience recurrent bouts of epilepsy, a chronic neurological illness marked by unexpected and frequent seizures. Skull fractures, genetic predispositions, tumors, and other contributing variables are among the many disorders that can cause epileptic seizures [2]. The condition affects 80% of individuals who reside in low- and middle-income nations. If these patients receive the right diagnosis and care, 70% of them can enjoy seizure-free lives. These patients are three times more likely to die young than the overall population. Three-quarters of epileptic patients in underdeveloped nations are unable to pay for care [3]. For the diagnosis of epileptic seizures, a number of techniques have been put forth thus far, including clinical and neuroimaging techniques. Physicians employ neuroimaging techniques extensively [4]. Neuroimaging techniques are generally divided into two categories: structural and functional. One functional neuroimaging technique for diagnosing epileptic seizures is electroencephalography (EEG) [4]. It takes a lot of time and effort for medical experts to visually analyze electroencephalograms (EEGs) and process the data. Automatic seizure diagnostic techniques that combine machine learning and signal processing have been launched as computer-aided diagnosis (CAD) systems in order to get around these restrictions.

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Seizures are classified as focal (originating in one brain area) or generalized (affecting both hemispheres), with EEG patterns showing rhythmic low-amplitude waves in healthy states, irregular disruptions near tumors, and high-frequency spikes during ictal events, aiding in precise detection. The scalp is equipped with electrodes to record EEG signals. The electrical impulses produced by brain neurons are detected by these electrodes. Noise and extraneous information are frequently included in raw EEG signal data. To improve the quality of the signals, pre-processing techniques like filtering, baseline correction, and artifact removal are used. Because of the relatively high complexity of EEG data, accurately diagnosing epileptic seizures can be tricky [2]. In addition to revealing information beyond independent characteristics, modeling intra-brain EEG dependencies is essential for enhancing the functionality of emotional brain-computer interfaces [5]. EEG's multi-band functional connectivity demonstrates that undesired cognitive states may be accurately identified by analyzing altered correlations between channels [6]. EEG-based deep architecture highlights the significance of extracting sensitive and multi-scale characteristics and may identify minor emotional changes [7]. Additionally, the potential of sophisticated deep learning models in epileptic seizure prediction has been shown utilizing Transformer and the extended attention mechanism [8]. Although these methods often lack the requisite interpretability and durability in dynamic clinical situations, combining EEG with hierarchical decision modeling might provide insightful information about cognitive processes [9]. However, because it may result in the creation of strongly correlated predictors, this strategy has limitations. When compared to other machine learning techniques, decision trees' high training costs represent another significant issue. This algorithm's greedy search strategy during the tree-building process is the cause of the issue. To lower this expense, the decision tree is trained using a fuzzy logic system, which in this study is a novel method from an adaptive fuzzy inference system for the tree construction process. This will lessen the issue of over fitting as well. However, we can create a feature map utilizing automatic encryption and a feature reduction matrix to create an ideal system that is resilient to data changes.

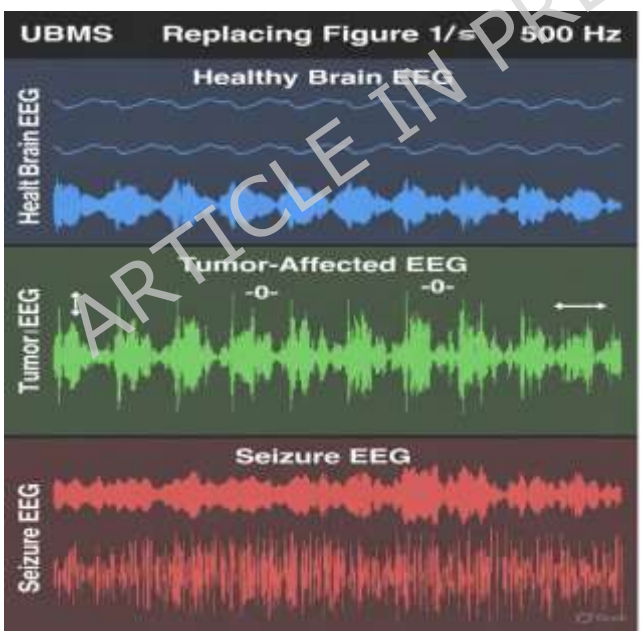

Figure 1: EEG signal changes in different brain states [10].

Seizures can be categorized into many varieties according to their features and the parts of the brain that cause them. We can observe the various patterns of EEG signals collected from the healthy brain area, the tumor-affected brain area, and during a seizure event in Figure 1. Regular, rhythmic patterns with consistent frequency and amplitude are typically observed in healthy brain regions, signifying normal electrical activity. The EEG signals exhibit alterations at the tumor site in comparison to the signals in the healthy brain region.

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Depending on the type and location of the tumor, these alterations may show up in a variety of ways. On the other hand, the EEG signals exhibit clear patterns during a seizure occurrence, indicating high frequency and amplitude aberrant neural activity. The EEG signal for a certain time period is displayed on the X-axis in the above image, while the signal amplitude is displayed on the Y-axis [13]. The many instances of EEG signal alterations depicted in Figure 1 are interpreted in Table 1.

Table 1- Present classes Of EEG signal [10].

Signal number	Description
1	Recording of seizure activity
2	Recording of the tumor location
3	Identify tumor location and EEG recording from healthy part of Brain
4	Eyes closed during recording
5	Eyes open during recording

The Glasgow Coma Scale (GCS) is used to gauge a coma patient's level of consciousness. Because it measures the patient's result multiple times during the day, the GCS is crucial for accurate and efficient patient assessment as well as for designing suitable treatment strategies and patient care. Complete consciousness is represented by a score of 15 while the lowest degree of consciousness, or coma, is represented by a score of 3. The GCS score in a coma is typically interpreted as follows [52]:

Score 3–8: severe or profound coma

Score: mild coma, 9–12.

Light anaesthesia or moderate coma, score 13 to 15.

Accordingly, the level of consciousness in a coma is typically lower than 8 [53].

The Glasgow Coma Scale is a worldwide tool used to determine an individual's IQ or state of consciousness. There are three distinct parts to this scale: voice, movement, and sight. There are four points for the eye region, five for speech, and six for movement. As a result, the most conscious condition, the normal state, receives a score of 15, while the least conscious state receives a score of 3. In most situations, the Glasgow Coma Scale is used to assess the extent of brain damage, but it can also be used to assess stroke, infections, seizures, and other conditions. Mild brain injury is indicated by scores between 13 and 15, moderate brain injury by scores between 9 and 12, and severe brain injury by scores of 8 or lower. The lower this number is, the more severe the injury, the worse the condition, and the higher the mortality rate [54, 55]. These requirements for deep coma stages of 3 to 8 for various classes 1-6 are defined in Table 2.

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Table 2. Glasgow Coma Criteria Table [52-55].

	1	2	3	4		5	6
Eye response	Not opening the eyes	Open your eyes in response to pain.	Open your eyes in response to the sound.	The eyes open spontaneously and without external stimulation.		N/A	N/A
Verbal response	No sound is produced.	Unintelligible voice	Using inappropriate words	Conversation that shows confusion.		Paying attention to the word, one must be fully aware (oriented) of time, place, and person.	N/A
Motor response	Complete limpness and	Decerebration (opening of	Decorticate (drawing	Distracting yourself from		Moving the pain agent	Carrying out the

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	immobility of the limbs	limbs) in response to pain	together of limbs) in response to pain	the pain factor		away from yourself (performing active movements to move the pain agent away)	examiner's command to move different parts of the body

In this study, we propose a novel hybrid machine learning model for epileptic seizure detection, named DT-FIS, to overcome the aforementioned difficulties. The model uses a decision tree-based learning system in conjunction with a methodical adaptive fuzzy inference methodology for output classification. The input of the model is a feature vector that has been derived from raw EEG signals using time-frequency transformation and a variety of statistical techniques. Based on the output features in each cluster, an automatic cryptographic system optimized with the water cycle algorithm (AE-WCA) is then used to decrease the number of features and raise the density and density of samples. The accuracy of clustering various sample classes is improved by this procedure. The categorization process is now carried out using the suggested machine learning system with the aid of optimized and reduced data. Using the starfish optimization approach, this hybrid DT-FIS machine learning system is trained. The issues with this approach are avoided by this one, which is based on the suggested decision tree approach. Despite EEG data's instability, high variability, and nonlinearity, the suggested fuzzy classifiers may be helpful for analysis. For big datasets with simpler models and the information overhead problem, fuzzy classifiers are more computationally efficient than decision tree classifiers. However, they can still achieve acceptable accuracy for specific data types, such as EEG signals. Additionally, despite the fact that the data is nonlinear, the suggested automatic coding system is able to identify linear relationships in it. This indicates that an AE-WCA may identify linear correlations between various aspects or signal components even though EEG signals may be nonlinear and have an entangled structure. The DT-FIS approach, on the other hand, is perfect for analyzing EEG signals because of its reputation for withstanding uncertainty in complicated and high-dimensional data. This model can be made more resilient and expressive by combining the best aspects of both designs. We evaluate the efficacy of this model using the CHB-MIT dataset and demonstrate that it performs better than the state-of-the-art methods for epileptic seizure identification at the moment. More accurate seizure detection could result from the suggested methodology, which could eventually improve the management and treatment of epilepsy.

The following are the main contributions of our work.

- We present DT-FIS, a hybrid machine learning model designed to automatically detect and categorize epileptic episodes.
- Using the starfish optimization algorithm to train the adaptive fuzzy inference system and decision tree, we design the model from the standpoint of time-frequency feature extraction in order to fully explore the temporal and spatial relationship between multiple EEG channels.
- We employ an automatic encryption strategy to improve the inter-cluster discrimination for the training samples while simultaneously decreasing the features and boosting the density of samples in each class. The water cycle algorithm is used to optimize this system.
- We perform several tests on the UBMC dataset. The results show that the proposed strategy outperforms the compared strategies and provides academics and physicians with new ideas.

This is how the remainder of the paper is organized. The relevant literature on diagnosing epileptic seizures is reviewed in Section 2. The pre-processing method and its steps, as well as the suggested seizure detection strategy, are the main topics of Section 3. The experimental setup and discussions based on the suggested code and module are presented in Section 4 after a quick review of the dataset used. The findings are analyzed, discussed, and contrasted with those of earlier research in Section 5. Lastly, the paper's conclusions are presented in the last part.

2. Related works

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There has been a lot of interest in using graph neural networks to take use of the implicit information in seizure detection. These networks consist of interacting nodes linked by edges whose weights are based on anatomical connections or temporal correlations. To get over this restriction, a novel hybrid framework for epileptic seizure detection that combines DenseNet and linear graph convolutional neural networks (LGCN) is presented in [11]. By enhancing feature propagation in each of its layers and decreasing the fading gradient issue, DenseNet outperforms earlier deep learning networks in terms of computational accuracy and memory efficiency. This hybrid framework surpasses the state-of-the-art in seizure detection, achieving 98% accuracy and 98.60% specificity in extensive tests on the public CHB-MIT EEG dataset. The raw EEG signal is preprocessed using the Stokkell transform (S-transform), and the resulting matrix is then grouped into time-frequency blocks as input to LGCN for feature selection, followed by DenseNet for classification.

A new deep learning-based technique for automating seizure identification utilizing log-Mel spectrograms of EEG signals is presented in Paper [12]. By applying logarithms to the signals' Mel-filter bank spectrograms, this technique creatively converts EEG data into visuals, enabling convolutional neural networks (CNNs) to categorize the resulting image representations. These pictures are used to train a CNN model that is intended to categorize EEG signals. Two publicly accessible datasets one for binary classification and the other for multi-class classification are used to assess the suggested approach. The five-class classification accuracy, sensitivity, specificity, precision, and F1 score of the suggested method are 98.13%, 95.33%, 98.83%, 95.59%, and 95.32%, respectively, according to experimental results on the Bin dataset. 99.60%, 99.33%, 99.67%, 99.52%, and 99.41% are the accuracy, sensitivity, specificity, precision, and F1 score for three-class classification on the same dataset, respectively. The NSC-ND dataset's three-class classification yields the following results: 93.33%, 90.00%, 95.00%, 91.89%, and 89.50% for accuracy, sensitivity, specificity, and precision, respectively. These results reveal the possibility for automatic epilepsy diagnosis in the actual world and show how well log-Mel spectrograms work with CNNs for EEG-based epileptic seizure identification.

Paper [13] introduces a novel method for detecting epileptic seizures that makes use of the advantages of deep learning (DL) and machine learning (ML) algorithms in EEG data. Neurological occurrences known as epileptic seizures have garnered significant recognition among academics due to their unique properties found in electroencephalograms (EEG). Deep learning (DL) and machine learning (ML) algorithms have become effective tools for classifying and extracting features from EEG signals. Numerous researches have classified EEG signals, computed time-frequency domain characteristics, and/or converted them into pictures. This work focuses on using deep learning-based one-dimensional convolutional neural network (1D CNN) techniques and machine learning-based classifiers to classify time series data representations of EEG signals by adjusting parameters. In addition to identifying the best classifier, the primary goal is to highlight crucial parameters like sensitivity, accuracy, and precision all of which are crucial for medical research, particularly for early disease diagnosis and patient care improvement. Time series data points taken from EEG signals make up the UCI epileptic seizure detection dataset used in this investigation. After preprocessing, the dataset was input into classifiers such as TabNet, Random Forest (RF), Extreme Gradient Boosting (XGBoost), and One-dimensional Convolutional Neural Network. These classifiers achieved encouraging accuracies of 98%, 96%, 98%, and 99%, respectively. The accuracy, sensitivity, precision, and recall of the suggested 1D-CNN model were all better than those of other cutting-edge models.

HyEpiSeiD, a resilient deep learning framework, was proposed in [14]. It uses a convolutional neural network and two gated recurrent unit layers to extract self-learning characteristics from pre-processed EEG data and makes predictions based on those features. Two publicly available datasets, the Mendeley and UCI Epilepsy datasets, are used to assess the suggested HyEpiSeiD framework. The suggested HyEpiSeiD model outperformed the majority of state-of-the-art techniques in the diagnosis of epilepsy, with classification accuracies of 99.01% and 97.50%, respectively.

A comparative comparison of publications devoted to automatic epileptic seizure detection using machine learning and deep learning approaches is conducted in [15]. The UCI-Epileptic Seizure Recognition dataset is used for training and validation. To determine the optimal strategy, a suggested model that makes use of long short-term memory (LSTM) is combined with certain traditional ML and DL techniques. The

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optimal method for detecting epileptic seizures is then determined by comparing various algorithms. When compared to the other algorithms discussed in this study, the suggested LSTM model yields the most appropriate and correct outcome, with a validation accuracy of 97%.

An adaptive control framework using fuzzy systems and a hierarchical structure for unknown nonlinear systems is introduced in [16], employing an event-triggered method to alleviate processing demands. The integration of adaptive fuzzy logic with hierarchical architecture facilitates the management of uncertainty and intricate system dynamics. This study primarily addresses control issues, however the use of multi-level fuzzy structures and adaptive learning is theoretically consistent with hybrid fuzzy intelligence methodologies employed in data-driven systems.

The work [17] presents an enhanced reinforcement learning approach using the particle swarm optimization (PSO) algorithm for the effective management of nonlinear systems with saturated inputs. The proposed approach integrates reinforcement learning with meta-heuristic optimization to derive optimum solutions amid uncertainty and practical restrictions. The concept of using Particle Swarm Optimization (PSO) to adjust learning parameters serves, from an algorithmic standpoint, as a motivation for utilizing meta-heuristic approaches in the optimization of intricate machine learning models.

In [18], researchers introduced a graph autoencoder model for inferring gene regulatory networks from single-cell data, which operates by acquiring a compact and organized representation of high-dimensional biological data. This method demonstrates that autoencoders may uncover latent relationships and intricate network topologies without requiring explicit labels. These findings fundamentally reinforce the significance of AE-based approaches in dimensionality reduction and the extraction of significant features from intricate biological data.

The research [19] introduces a machine learning framework for detecting microRNA–disease connections via low-order approximation, link propagation, and multikernel learning techniques. The model integrates low-rank modeling with multiple kernel learning to elucidate intricate and nonlinear interactions between biological constituents and illnesses with enhanced precision. This methodology exemplifies the effective use of sophisticated machine learning techniques in disease modeling and underscores the significance of compact representations and kernel learning in diagnostic challenges.

The research [20] suggests a sophisticated machine learning-based epileptic seizure detection algorithm to identify EEG signals with high classification accuracy rates. The improved adaptive neuro-fuzzy inference system (IANFIS)-light gradient boosting machine (LightGBM) hybrid technique is presented in this paper for the automatic diagnosis and detection of epilepsy from EEG data. EEG records from the University of Bonn in Germany and scalp EEG data from Boston Children's Hospital corroborated the trial results. The results show that in both cases, the suggested IANFIS-LightGBM has the best classification accuracy ratings.

To predict epileptic seizures, the research [21] suggests an optimization-controlled hybrid group classifier that combines the AdaBoost, random forest (RF), and decision tree (DT) classifiers for automatic analysis of EEG signal datasets. To prepare the EEG signal for feature selection, it is first preprocessed. Alpha, beta, delta, theta, and gamma wave data from EEG are sent to the feature selection process. The suggested hybrid search optimization technique extracts significant features, including statistical features, wavelet features, and entropy-based features. After these features are extracted, the suggested group classifier generates the expected result. The suggested hybrid search optimization method is created by fusing the characteristics of gregarious and corvid search agents, and it is applied to assess the group classifier's integration parameters. For the CHB-MIT database, the suggested technique's accuracy, sensitivity, and specificity are found to be 96.6120%, 94.6736%, and 91.3684%, respectively. This demonstrates how well the suggested method for early seizure prediction works. For the Siena Scalp database, the suggested technique's accuracy, sensitivity, and specificity are 95.3090%, 93.1766%, and 90.0654%, respectively, demonstrating its efficacy in the early seizure prediction process.

3. Proposed classification techniques and methods

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In order to train and improve the functionality of different system components, this study has used two optimization techniques: Goose and Grey Wolf. Furthermore, the SVM-Fuzzy hybrid machine learning system and feature reduction matrix have been used to implement feature reduction techniques. In this part, we will review each before outlining the recommended strategy.

3.1. Overview of the Starfish Optimization Algorithm (SFOA)

Inspired by the behaviour and biological traits of starfish in marine habitats, the Starfish Optimization Algorithm is a new algorithm in the field of nature-based optimization. The program specifically looks at how starfish may adapt to their surroundings, repair injured arms, and look for food on the ocean floor. Its primary method relies on the random movement and group collaboration of starfish to locate the best locations inside the search space. This method is appropriate for tackling complicated multivariate issues because it strikes a balance between exploration (identifying new regions) and exploitation (enhancing current solutions) through the use of nonlinear movement patterns and adaptive mechanisms. This algorithm views the starting population as a collection of starfish that use basic mathematical functions to adjust their positions in response to their closeness to food sources (optimal solutions). In order to prevent local optima and boost search variety, the regeneration and bioplasticity mechanisms of starfish have been modelled.

Similar to search, hunting, and regeneration behaviors, SFOA includes periods of exploration and exploitation [22]. For five-dimensional or one-dimensional searches, depending on the problem dimension ($d > 5$ or $d < 5$), SFOA employs a five-arm structure (with eyes on arms), in contrast to common algorithms that rely on vector-based search for separable problems. Convergence concerns in non-separable situations are addressed here. Hunting and regeneration tactics are used during the exploitation phase. Hunting update solutions use data from two starfish to direct candidates toward more advantageous places [23]. Figure 2 shows the flowchart for SFOA. Compared to other meta-heuristic algorithms, this algorithm's multidimensionality for problem-solving and searching makes it an excellent choice for training the suggested machine learning system.

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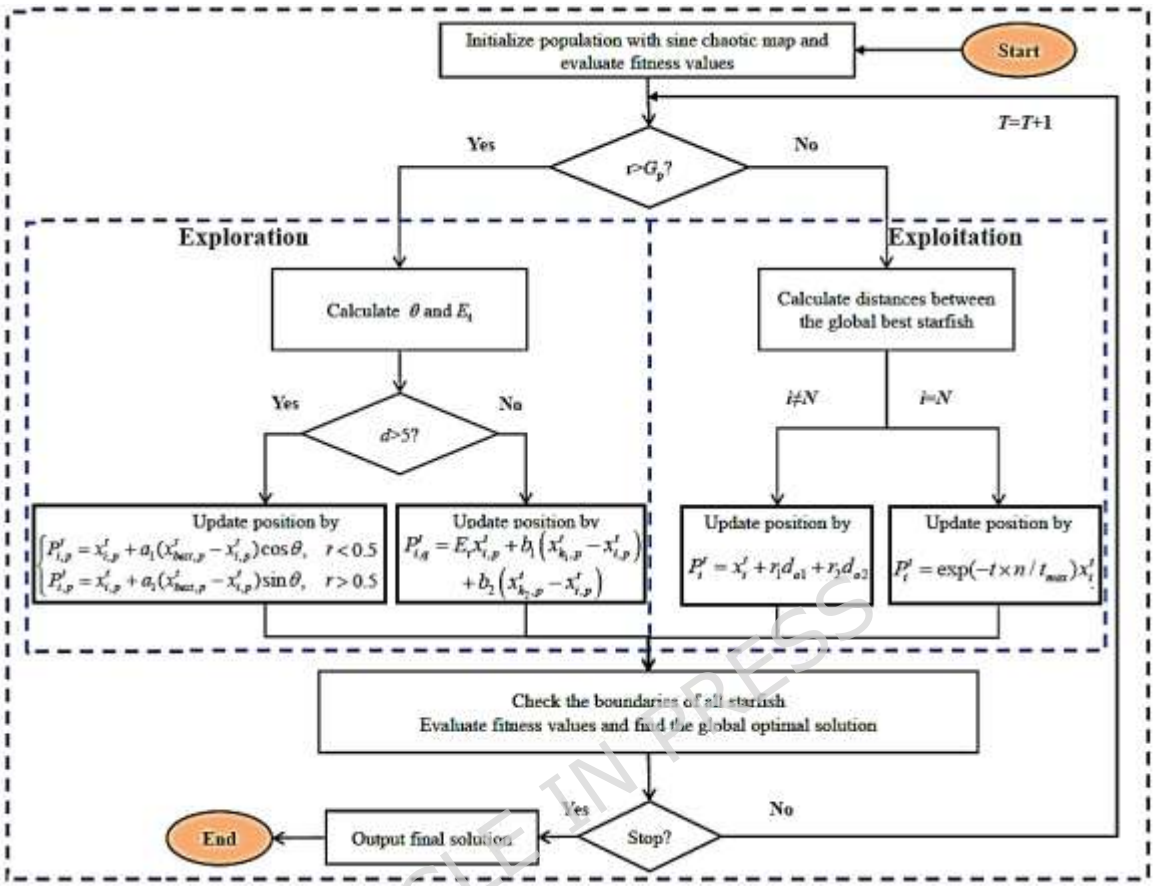

Figure 2. SFOA Flowchart [23].

3.2. Water Cycle Algorithm (WCA)

In 2012, Eskander et al. presented the Water Cycle Algorithm (WCA), a nature-inspired optimization technique intended to address challenging and nonlinear optimization issues. The natural hydrological water cycle, which involves evaporation, precipitation, and the movement of water from rivers to lakes and seas, served as the model for the algorithm. The initial population in this method is thought of as streams that flow toward ideal locations, much like lakes and seas. The algorithm's primary method is to employ basic mathematical formulas to move streams in the direction of the best answers, and it uses the chance of evaporation to diversify the search. The technique is useful to a range of challenges, such as resource management and engineering optimization, because it strikes a balance between exploring new regions and enhancing old solutions. An initial population is created at random during the algorithm execution process, and the top solutions are chosen to serve as the primary rivers and seas. After that, a motion relation determined by the distance and direction to the oceans is used to gradually update the streams. To prevent local optimization, the evaporation process is triggered with a specific probability, resulting in precipitation at random locations or near the seas. This process keeps going until a stopping requirement is satisfied, like completing a specific number of iterations or obtaining the required accuracy. Future study may find the WCA algorithm to be a good option for tasks like automatic feature encoding and feature reduction matrix parameter optimization due to its ease of use and strong capacity to solve problems involving several variables. The flow chart for this algorithm's execution is displayed in Figure 3 [42].

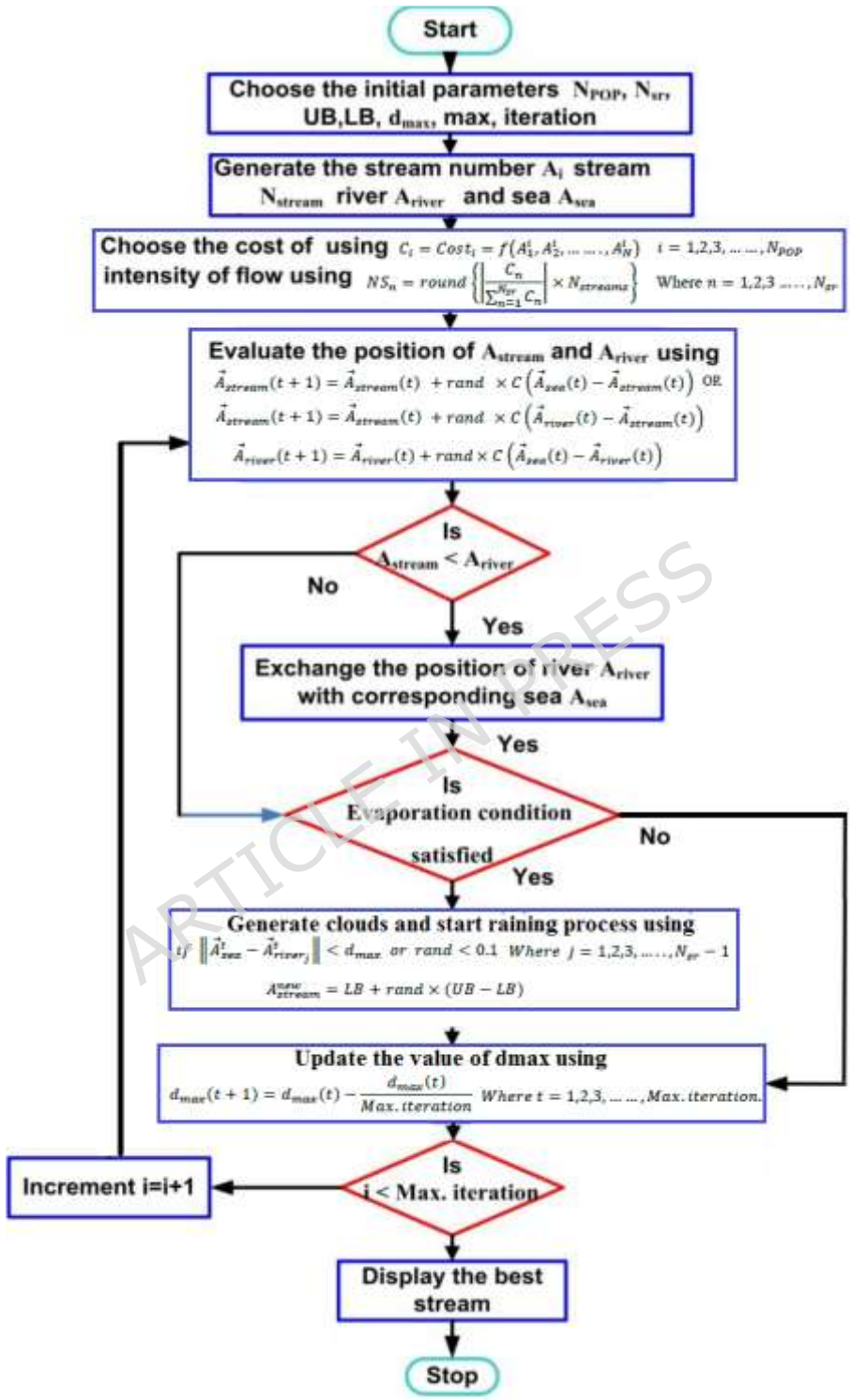

Figure 3. WCA Flowchart [42].

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3.3. Steps to diagnose epilepsy seizures in the proposed system

There are various components to CAD's procedures for detecting epileptic seizures. In order to extract various features from individual EEG signal channels, the incoming EEG data is first prepared after filtering and pre-processing techniques. The classification system's feature vector dimensions and complexity are then decreased with the use of the feature reduction matrix. Ultimately, the suggested machine learning method transforms the encoded data in the classification stage into distinct types of epileptic seizures. The algorithms suggested in the preceding sections are used to solve the suggested problem. The remainder of this section goes into detail about each component. The system steps' block diagram is displayed in Figure 4.

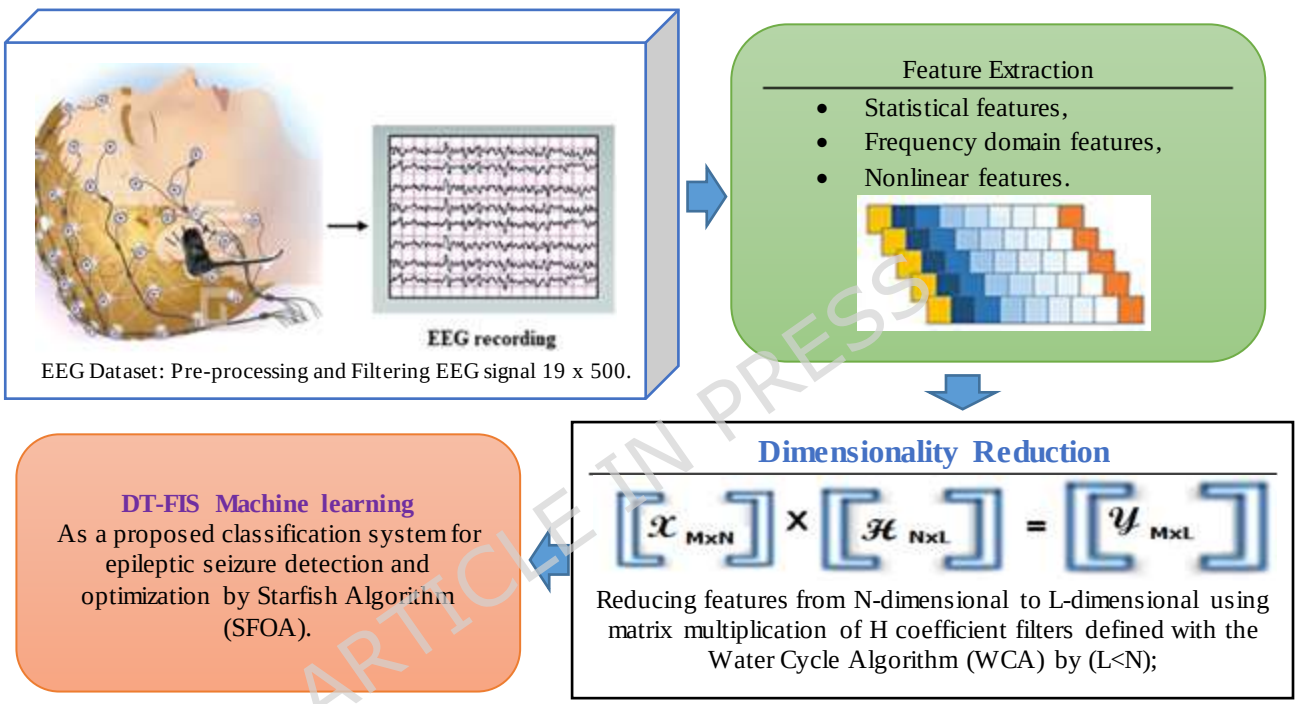

Figure 4. Block diagram of the proposed scheme for detection and classification.

The steps of the suggested method for identifying and categorizing epileptic episodes using EEG signals are shown in detail in Figure 4. The EEG signals, which have dimensions of 19 × 500, are first preprocessed and filtered before features, such as statistical, frequency domain, and nonlinear features, are extracted. Next, matrix filter coefficients optimized by the Water Cycle Algorithm (WCA) (L < N) are used to reduce dimensionality from N to L. In order to identify epileptic seizures, the Starfish Algorithm (SFOA) optimizes the DT-FIS (Decision Tree-Based Fuzzy Network) classification system.

3.3.1. Feature extraction

This section identifies epileptic events in EEG records using a variety of feature extraction approaches. Feature extraction methods for EEG signals include frequency domain, nonlinear, and statistical features. Nonlinear features based on FD provide useful information about EEG data. EEG waves exhibit chaotic behavior. The FD approach is one of the nonlinear strategies that can be used to extract important information from EEG data. Each of these methods is discussed in the next section.

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3.3.1.1. Statistical features

The most significant statistical features are chosen as indicated in Table 3 [26] and are used to extract meaningful signal information.

Table 3. Statistical characteristics to diagnose epilepsy seizures [26]

Formula	Feature Name
$\boldsymbol{X_{mean}} = \frac{1}{n}\sum_{1}^{n} x_i$	Mean
$\boldsymbol{X_{var}} = \sum_{n=1}^{N}(x_n - AM)\frac{2}{N-1}$	Variance
$\boldsymbol{X_{ku}} = \sum_{n=1}^{N}(x_n - AM)\frac{4}{(N-1)SD^4}$	Kurtosis
$\boldsymbol{X_{Ske}} = \sum_{n=1}^{N}(x_n - AM)\frac{3}{(N-1)SD^3}$	Skewness
$\boldsymbol{X_{std}} = \sqrt{\sum_{n=1}^{N}(x_n - AM)\frac{2}{n-1}}$	Standard Deviation
$Max(x_n)$	Max

3.3.1.2. Frequency characteristics

1) Intensity Weighted Mean Frequency (IWMF)

The intensity weighted mean frequency (IWMF), sometimes referred to as the mean frequency, is obtained by multiplying the normalized power spectral density (PSD) by the frequency. Paper [27] uses x[k], the normalized PSD of the signal period at frequency f[k], to calculate the IWMF.

$$\boldsymbol{IWMF(x)} = \sum_{\boldsymbol{k}} \boldsymbol{x[k]f[k]} \qquad (\mathbf{1})$$

2) Intensity Weighted Bandwidth (IWBW)

The PSD width and frequency-weighted standard deviation measures are available from [27].

$$\boldsymbol{IWBW(x)} = \sqrt{\sum_{\boldsymbol{k}} \boldsymbol{x[k](f[k] - IMWF(x))^2}} \qquad (\mathbf{2})$$

A substantial change in the PSD results in a reduced IWBW, where x[k] is the normalized PSD and IMWF is the average frequency of the input signal period.

3.3.1.3. Power spectral density

The Power Spectral Density (PSD) is a very effective technique for signal analysis in the frequency domain. Several properties, including spectral edge frequency (SEF), intensity-weighted bandwidth (IWBW), intensity-based mean frequency (IWMF), and other parameters, can be derived from the PSD and its normalized form. Fuzzy models that are robust to noise, uncertainty, and EEG variability are in line with fault-tolerant learning-based control under uncertainty frameworks for nonlinear systems with uncertain communication channels [27]. The use of this feature extraction method to regulate cursor movement in brain-computer interface (BCI) systems has been examined in a case study in this area. The

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outcomes demonstrated that PSD-based approaches outperform earlier cursor movement strategies in terms of accuracy. For uncertain contexts, event-triggered adaptive learning has been created. This concept is consistent with the post-AE-WCA strategy of decreasing data overhead, concentrating on efficient information samples, and raising the information density of features [28].

3.3.1.4. Band power

EEG-BCI is also frequently used for wheelchair control. For each of these systems to function, a feature extraction phase is required. In one study, the total band power (BP) of steady-state visual evoked potentials (SSVEPs) was used to detect stimulus frequencies. The band power for each stimulus frequency is computed as follows [29].

$$\widehat{P}_{k,l} = \left\| X_k^T s_l \right\|^2, \qquad (3)$$

where sl is the second channel signal and X is a noise-free SSVEP model.

A linear classifier is used to classify the frequency on which the topic is focused after the power has been estimated. The minimum accuracy of this system when moving a wheelchair was 93.61%. Furthermore, it was found that stress-inducing circumstances did not significantly affect the subject's performance. How are decision stability maintained by intelligent models that are resilient to disruptions and structural alterations. By decreasing spurious correlation, addressing signal instability, and improving generalization capability, DT-FIS precisely addresses this issue in the EEG space [29]. For multi-agent systems that are prone to errors and data breaches, adaptive neural event-triggered secure learning has been created. The use of automatic encoding (AE) as a layer for protecting, enhancing, and stabilizing EEG data before fuzzy categorization is thus directly supported theoretically [30].

3.3.1.5. Entropy properties

This article uses a range of entropies to derive the characteristics of EEG signals. The entropy-based features show anomalies in the signal and are more resilient to noise than earlier methods. Entropy-based signal uncertainty metrics offer reliable indicators of anomalous dynamic behaviors under drift and actuator defects, as recent adaptive fault-tolerant modeling has shown [31]. In order to capture hidden complexity fluctuations in non-stationary EEG signals and enable robust identification between normal, tumorous, seizure, and coma brain states, this study employs Shannon entropy and related entropy descriptors. An example of the entropy relationship is shown below.

(1) Shannon's feature

This entropy was proposed by reference:

$$E_{Sh} = -\sum_{n=1}^{x} S_n \log_2 S_n \qquad (4)$$

In the equation (4), S_n is probability of the value of the feature.

(2) Log-energy entropy

Also referred to as [30,31], the log-energy entropy is a measure of the complex intensity of signals.

$$E_{Log} = \sum_{i=o}^{K} \log\left(E_i^2\right) \qquad (5)$$

(3) Shannon Violet's entropy

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In this section, the average of Shannon's waveguard entropy is displayed. Multi-band wavelet energy distribution has been shown to offer reliable indicators of hidden dynamic disturbances and anomalous system behaviors in extended state observer-based nonlinear modeling [33]. Inspired by this idea, this study uses Shannon Violet's wavelet entropy to measure EEG energy redistribution across frequency bands, allowing for noise-resilient seizure and coma state detection. The total signal energy can be written below if ET shows the energy under the first band, which is controlled by the wave coefficients:

$$\boldsymbol{E_t = \sum_{i=1}^{K} E_i} \qquad (\mathbf{6})$$

With K being the total number of EEG signals obtained from the wavelet subbands, the wavelet energy can be calculated as follows [46].

$$q_i = \frac{E_i}{E_t} \qquad (7)$$

The Shannon-based wavelet entropy relation is defined as follows [48]:

$$\boldsymbol{Swn = -\sum_{i=1}^{K} q_i \log(q_i)} \qquad (\mathbf{8})$$

Finally, the mean Shannon wavelet entropy is constructed as follows [46] using swn_x and swn_y, which are the representations of the x and y time series Swn of the EEG signal:

$$swn_{avg} = \frac{swn_x + swn_y}{2} \qquad (9)$$

(4) Average Entropy of Rényi Wavelet

The Entropy of the Rényi wavelet is defined in equation (10) [35]:

$$Rwn_\alpha = \frac{1}{1-\alpha}\log\left(\sum_{i=1}^{K} q_i^\alpha\right), \quad \alpha \neq 1 \qquad (10)$$

In this instance, it is assumed that the value of the parameter "a" is 2. An alternative definition of the Rényi entropy is given by equation (11) [5]:

$$Rwn_2 = -\log\left(\sum_{i=1}^{K} q_i^2\right) \qquad (11)$$

The Rényi wavelet's average entropy is defined as follows, which is comparable to equation (9).

$$Rwn_{Avg} = \frac{Rwn_x + Rwn_y}{2} \qquad (12)$$

(5) Average TsallisWavelet Entropy

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In reference [34], the Tsallis wavelet entropy has been studied in detail. Tsallis-entropy-based wavelet energy descriptors may reliably capture hidden nonlinear structural changes under input dead-zones and disturbances, as event-triggered nonlinear inverse learning has shown [34]. Inspired by this idea, this study uses average Tsallis wavelet entropy to describe nonlinear EEG energy redistribution, allowing for reliable separation of profound coma and seizure states. The definition of the Tsallis wavelet entropy is as follows:

$$Twn_{\alpha} = \frac{1}{1-\alpha}\left(1 - \sum_{i=1}^{K} q_i^{\alpha}\right), \quad \alpha \neq 1 \qquad (13)$$

Where the non-extensibility index is denoted by the parameter a. The following formula is used to get the Tsallis wavelet's mean entropy [48]:

$$Twn_{Avg} = \frac{Twn_x + Twn_y}{2} \qquad (14)$$

3.3.1.6. Hjorth parameter properties [35]

Activity, mobility, and complexity are the three primary parameter categories that make up the Hjorth parameter, a technique for expressing the statistical characteristics of signals in the temporal domain (Table 4). The level of the power spectrum in the frequency domain is determined by the activity parameter, which is the variance of the signal's time function; if the signal contains high-frequency components, the value of this parameter rises or falls noticeably. Indicating the dispersion of the signal's power spectrum, the mobility parameter the square root of the ratio of the signal's variance to its first derivative acts as an index for analyzing signal fluctuations. Another measure of the signal's form resemblance to a pure sine wave is the complexity parameter; the closer the value is to 1, the more structurally similar the signal is to a simpler sine wave.

Table 4. HJORTH parameter [35]

Parameter	Notation
Activity	var(y(t))
Mobility	$\sqrt{\frac{var\,(\acute{y}(t))}{var\,(y(t))}}$
Complexity	$\frac{mobility\ (\acute{y}(t))}{mobility\ (y(t))}$

These three factors offer helpful information on the frequency spectrum of signals in addition to aiding in time domain analysis. Utilizing these factors can also lessen the computing load and improve the efficiency of the analytical process.

PSD captures frequency content critical for identifying seizure-related bands (e.g., theta spikes); entropy reflects signal irregularities indicative of chaotic ictal activity; Hjorth parameters summarize temporal dynamics like mobility for detecting abrupt changes during seizures. Higher PSD in beta bands indicates seizure hyperactivity; elevated entropy suggests increased irregularity during ictal phases; higher Hjorth activity/complexity reflects amplified amplitude and frequency variations, signaling seizure onset. Combining these features creates a holistic representation, where PSD provides spectral context, entropy adds chaos measures, and Hjorth offers temporal insights, synergistically improving DT-FIS classification by capturing multifaceted EEG dynamics beyond isolated metrics.

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3.3.2. Features reduction

The best methods for lowering the computational complexity of these features for computational processes are feature dimensionality reduction approaches. One popular method for lowering feature dimensionality is to eliminate low-impact features from the classification discussion. The underlying structure and hidden constants of these low-impact features can have a significant impact on the classification accuracy of algorithms. In order to determine which feature is best suited for our work, we will employ an approach in this study that is influenced by all of the attributes. Nonetheless, the results of these features are displayed in the classification output in a manner that achieves very high classification accuracy while reducing processing time and complexity.

Reducing features from N=34 to L=10 enhances real-time applicability by lowering computational load (e.g., faster inference on edge devices) and improves generalization by mitigating over fitting in noisy EEG data, as shown in our ablation studies. [36] demonstrate that efficient feature compression techniques are necessary for brain analysis based on high-dimensional biological data since the existence of low-dimensional yet structural components can greatly impact the precision of diagnostic models. Mapping biological components to low-dimensional feature spaces improves the capacity to identify diseased states and decreases spurious correlations, according to dynamic modeling of epileptic neural networks in [37]. A noise-resistant fuzzy neural network is presented in [38], demonstrating how lowering the dimensionality of features in noisy nonlinear systems boosts learning stability and enhances generalizability. Clinical evidence based on fNIRS in [39] demonstrates that it is possible to accurately discriminate between different levels of cognitive impairment by extracting optimal subsets of brain properties while maintaining important information. Additionally, [40] presents a biosequencing machine framework that reduces computing overhead and improves classification accuracy by learning how to translate high-dimensional information to a compact space.

To lower the feature vector's dimensionality, a number of methods have been devised. These methods include unsupervised learning algorithms like PCA, AEs and a collection of unsupervised neural networks, as well as supervised learning algorithms like Fisher.With the use of a feature reduction matrix, the block diagram of the automatic feature vector encryption system is displayed in Figure 5. Two goals are taken into consideration when performing the vector multiplication between the optimized feature matrix and the feature vector in this study. The initial goal is to transfer features of length N to features of length L by appropriately training the feature reduction matrix as a dimensionality reducer of the feature vector space. Assuming N>L, the feature dimensionality reduction is carried out. The second goal is to use the feature mapping that has been developed to provide a high discrimination between features of various clusters and a high correlation of features in each class. A multi-objective functional model is employed to accomplish this goal. The proposed equations for this part focus the training EEG signal samples at the cluster centre and generate features with higher density and high inter-cluster distance in each cluster. An illustration of the functions for the two-cluster model is provided in Figure 5.

The Water Cycle Algorithm (WCA) approach is used as the optimizer to train this matrix reducer, and the distance parameter is minimized as a multi-objective function. All data are normalized in order to adjust the results to the most recent clustering information. The equations for the multi-objective function in Figure 5 of this suggested AE system are presented below.

The distance between nodes is a crucial component in these equations. If two m-dimensional points are x=(x1,x2,…,xm) and y=(y1,y2,…,ym), the dist function in R may compute a variety of distances. However, the Canberra distance was used in this work to calculate the separation between the feature variables. The following formula can be used to get this distance [41,43]:

$$d_{Canb} = \sum_{i=1}^{m} \frac{|x_i - y_i|}{|x_i| + |y_i|} \qquad (15)$$

The Euclidean distance type is another relationship that is used to compute the distance; it is derived from the relationship that follows [43]:

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$$d_{EU} = \left(\sum_{i=1}^{m} |x_i - y_i|^2 \right)^{0.5} \qquad (16)$$

The block diagram view of the feature reduction scheme utilizing the feature reduction matrix's feature multiplication is displayed in Figure 5.

This flowchart shows how to determine the weight parameters (W) in a feature dimensionality reduction matrix (N×L) using the Water Cycle Algorithm (WCA) technique. The main goal is to improve the quality of data clustering by applying two criteria:

- Lowering the intra-cluster dispersion metric D, which takes into account the average distance between clusters in each class or the maximum distance between cluster nodes and the cluster center of gravity.
- Optimizing the value of K, this might be a clustering quality metric like intra-cluster similarity or an index like Silhouette Score. In this paper, the separation between cluster centers of gravity is presented.
- To draw a line across clusters in the characteristics of various samples, the value of H must be decreased. The more precisely the cluster boundaries are defined, the lower this value is.

The objective function for defining the feature dimension reduction matrix is displayed in pseudo code 1.

We now compute the objective functions for a 2-cluster sample using the model displayed in Figure 6. Three functions are taken into consideration in this work. The average distance between each cluster's nodes and center of gravity is shown by the first function. This function, which is introduced with D in Figure 6, is expressed by Equation 17. In these relations, $X_{i,j}$, represents the position of the feature vector of each sample, S_j is the center of gravity of the clusters j, m_j is the number of samples in each cluster, and N is the number of clusters.

$$DD = \frac{1}{N} \times \sum_{j=1}^{N} d_{canb}(x_{i,j}, s_{i,j}, m_j) \qquad (17)$$

The second function, denoted by the letter K in Figure 6, is the minimum distance between the clusters' centers of gravity. Equation 18 defines this function as a matrix for the number of clusters greater than two.

$$KK = \min(d_{canb}(s_{i,j}, s_{i,j}, 1)) \qquad (18)$$

The third function, represented by the letter H in Figure 6 and computed using Equation 19, is also defined in the definition of the sum of the minimum and maximum distances of the features in each cluster.

$$HH = sum\,(d_{canb}(\max(x_{i,j}), \min(x_{i,j}), N)) \qquad (19)$$

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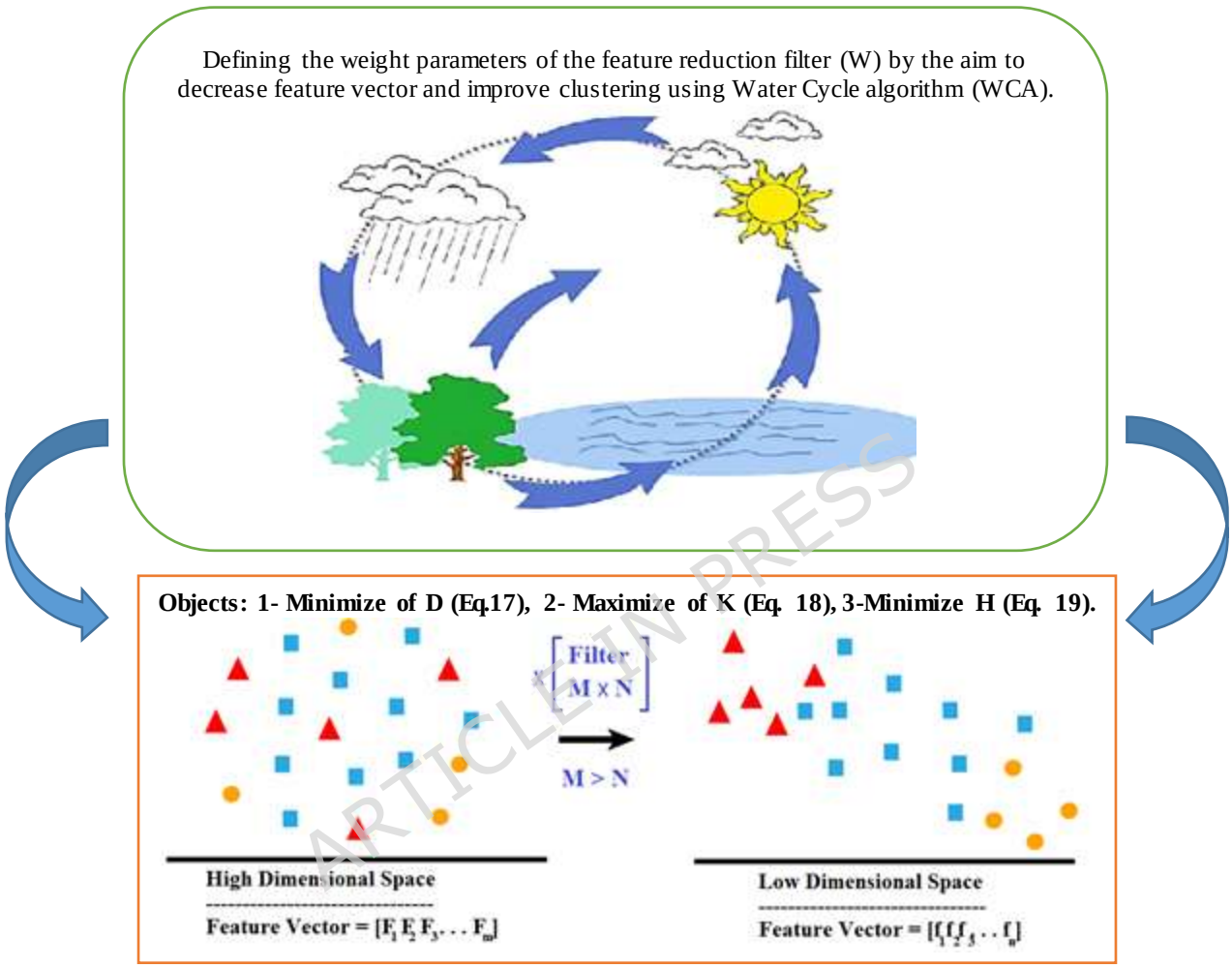

Figure 5. Showing the block diagram of the feature dimension reduction matrix system.

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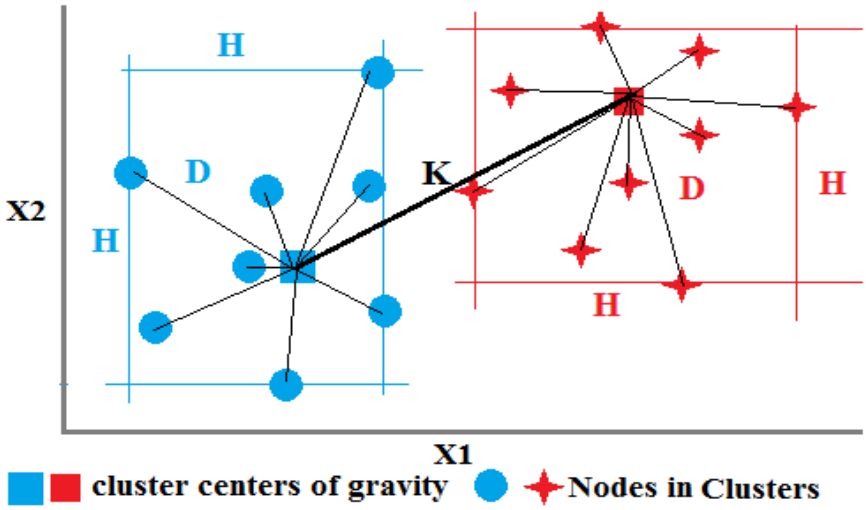

Figure 6. Sample of two clustering model.

Pseudo code 1: Objective function for calculating the feature reduction weighting parameter.

```
function [output] = fitnessfunction(w,X,Y,m,n)
W = reshape(w,m,n);
[D,K,H,YY] = featreduction(W,X,Y);
%;/
output = sum(D)/min(K(:))+0.1*sum(H);
end
-------------------------------------------------------------
function [D,K,H,Y] = featreduction(W,x,y)
% W : wight
% x : feature
y = y+1;
Y = abs(x*W);
D = zeros(1,max(y));
for i=1:max(y)
    S = Y(find(y==i),:);
    Xm(i,:) = mean(S);
    D(i) = mean(sum((abs((Xm(i,:) - S)./(abs(Xm(i,:))+abs(S))))'));
    H(i) = sqrt(sum((max(S)-min(S)).^2));
end
K = inf*ones((max(y)-1),max(y));
for j=1:(max(y)-1)
    for i=(j+1):max(y)
        K(j,i) = sum(abs(Xm(j,:)-Xm(i,:))./(abs(Xm(j,:))+abs(Xm(i,:))));
    end
end
end
```

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Optimized weights (W) are used to reduce the dimensionality of the data in order to enhance the more relevant features for clustering. This process enhances the precision and comprehensibility of clustering models (such as K-Means or DBSCAN).

3.3.3. Proposed classification system

The decision tree with adaptive fuzzy inference system (DT-FIS) network hybrid machine learning system will be used to detect epileptic seizures based on the features that were decreased in the previous step. This approach determines the class level in the output class using architecture akin to ANFIS. However, a Takagi-Sugeno type fuzzy logic system is used to compare each input feature as a binary value between 0 and 1. This section is based on a binary decision tree learning system paradigm. The block diagram of the suggested learning system is displayed in Figure 7, which we shall describe later.

Architecture of ANFIS

An adaptive framework that promotes learning and adaptation incorporates the Takagi Sugeno fuzzy method, or ANFIS. It is believed that a first-order Sugeno framework fuzzy model with IF-THEN rules provides the ANFIS design [44].

Rule 1: If x is A1 and y is B1, then f1 = p1x + q1y + r1

Rule 2: If x is A2 and y is B2, then f2 = p2x + q2y + r2

The inputs are x and y, and the fuzzy sets are Bi and Ai. The outputs of the fuzzy rule are represented by fi, whereas the design parameters of the training set are pi, qi, and ri. Each signal that the node receives is added up. The output is often shown as follows [44].

$$overall output = \sum_i (\overline{\omega_i}\ f_i) = \frac{\sum_i \omega_i f_i}{\sum_i \omega_i} \qquad (20)$$

As previously proposed, fuzzy c-means clustering (FCM) is likely the most effective method for implementing fuzzy inference system (FIS) in ANFIS. Backpropagation or hybrid approaches are the two methods available for training ANFIS. These methods directly affect FIS's inputs, outputs, and membership functions during the training process. Here, we have used goose optimization techniques to improve ANFIS's performance. In the proposed ANFIS, the parameters of the input membership functions (in this case, Gaussian) are first input into a vector. The goose optimizer is then used to select the ideal values in order to minimize a selected cost function. The cost function is as follows [44]:

$$\min_{\theta} Error = \frac{1}{N}\sum_{i=1}^{n} e_i^2 \qquad (21)$$

$$e_i = t_i - f(x_i|\theta)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(t_i - y_i)^2} \qquad (22)$$

where N is the number of ANFIS inputs, yi is the ANFIS output, n is the number of data samples, θ is the ANFIS parameters, xi is the input values, and ei is the error. Finally, the goose method is used to minimize the error. The goose algorithms are described in the sections above.

Architecture of decision tree

In machine learning, the Decision Tree Machine Learning System is a supervised technique for regression and classification. The data is arranged using this technique as a tree structure, with each internal node representing a choice based on a particular attribute, branches defining the decision

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outcomes, and leaves offering the final outputs (predicted value or categorization). Using metrics like the Gina Index or Information Gain, the learning process starts by choosing characteristics that offer the optimal data separation. Because of its ease of use, interpretability, and compatibility with raw data, this approach is frequently employed in domains including illness diagnosis and consumer behavior prediction [45].

To create an ideal model, a decision tree analyzes training data. Each step is carried out by choosing the branch that reduces data heterogeneity the most, for example, by using the Gina index or entropy. In order to make a forecast, fresh input is sent from the root to the leaves, and a certain route is taken to arrive at the outcome depending on the feature values. Nevertheless, this approach could result in overfitting, which can be avoided by using pruning approaches or by combining them with Random Forests [46]. We provide a straightforward numerical example of data mining with a decision tree to categorize information taken from EEG signals for the diagnosis of epileptic seizures in order to better demonstrate this learning system. Assume that the objective is to forecast whether or not a seizure will occur, and that the extracted features comprise the "Mean Frequency (MF)" and "Standard Deviation (SD)" of the EEG signals. The example's tree representation is displayed in Figure 7.

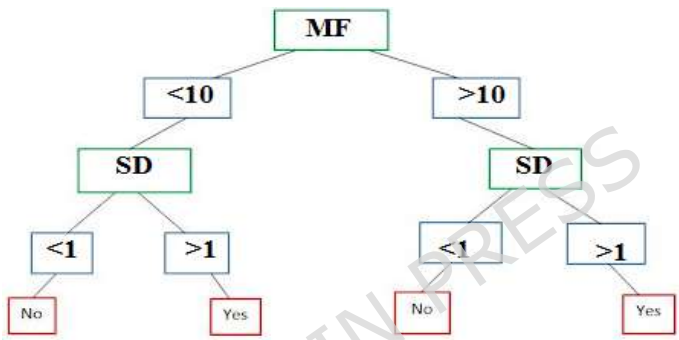

Figure 7. Decision tree for seizure detection from two features extracted from the EEG signal.

- (mean frequency = 6 Hz, standard deviation = 0.7) → No (no seizure)
- (mean frequency = 18 Hz, standard deviation = 1.5) → Yes (seizure)

According to Figure 7, the root has the characteristics of mean frequency with a threshold of 10 Hz and standard deviation with a threshold of 1. It concludes with conditional statements based on the branch shape and then moves on to the leaves with the outcomes of two states: "with seizure" and "without seizure." The decision tree in this example uses straightforward thresholds to classify the input. The output "yes" (seizure) is predicted for a signal with a mean frequency of 12 Hz and a standard deviation of 1.3. The path leads to the right node. By enlarging the data and modifying the thresholds, this data mining technique may identify epileptic episodes from EEG signals with greater accuracy [45]. This scheme's inability to achieve high accuracy as the number of input features rises, however, is consistent with the overfitting issue and prevents it from being trained to an ideal model for the system.

Architecture of DT-FIS

The two primary components of the suggested machine learning system are the branch classification or clustering section for classification and a feature comparison section for creating the tree's roots. The weights of each input feature will first be compared using a fuzzy logic system of the Takagi-Sugeno type. As illustrated in Figure 8, each feature's two parameters, ai and bi, add membership functions of types Z and S to the fuzzy logic system's input. A comparison border between 0 and 1 can be obtained by adjusting these two parameters. This system converts its output into a binary code that is either 1 or 0 for greater or lesser values. Regarding this, Figure 8 shows that we are given a binary output from which we can compute various output values between 0 and 2^m-1 using binary weighting between 2^L, where L is the number of the system's input characteristics. Actually, we used a binary to decimal conversion in this

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section. After that, we use an adaptive neural fuzzy system to normalize the output for the 0–1 range before moving on to the clustering stage. The decision tree approach labels the outputs of the preceding step in order to carry out the classification process. To determine these values, we will have to perform a fairly intricate search calculation as the number of input features rises. For instance, if there are just ten input characteristics, we will need to initialize 1024 distinct labels for the system and use meta-heuristic methods to carry out the required labeling for each of these values. For many input features, this search will make the process challenging. As a result, in this suggested approach, we will classify the output values of the prior class using a clustering pattern and an ANFIS adaptive neural fuzzy network. Gaussian membership functions will be used in this work to cluster and delimit the normalized values of the prior class, as illustrated in Figure 8. This work creates a Gaussian membership function for clustering the system's input between 0 and 1, with the parameters Ci, Di, and Classi standing for the center, radius, and label of each function, respectively. The number of search parameters rises with the number of Gaussian functions for clustering; for instance, in a system with 10 membership functions, we only need to initialize and search 30 parameters, which will reduce the information overhead.

The search parameters are initialized using SFOA, a novel meta-heuristic technique, in order to address this issue for the training dataset. To reduce the inaccuracy in figuring out the DT-FIS machine learning system's parameters and fuzzy membership functions, pseudo code 2 defines the goal function.

SFOA is explicitly integrated into the DT-FIS system to optimize fuzzy rules and tree depths, leveraging its bio-inspired regeneration mechanism to enhance convergence speed and accuracy in classifying nonlinear EEG patterns for epileptic seizure detection.

Pseudocode 2: Objective function to determine the parameters of the Gaussian membership functions and the SVM coefficients.

```
function [error] = DTFISfunction(x,in,out,fis1,fis2,Nin,NF)
%
A = [];
in(in<0)=0;
in(in>1)=1;
for i = 1:Nin
    fis1.Inputs.MembershipFunctions(1,   1).Parameters   =   [x(i)-
x(Nin+i)/2 x(i)+x(Nin+i)/2];
    fis1.Inputs.MembershipFunctions(1,   2).Parameters   =   [x(i)-
x(Nin+i)/2 x(i)+x(Nin+i)/2];
    IN(:,i) = evalfis(fis1,in(:,i));
    A = [A;2^(i-1)];
end
Input = IN*A;
for j = 1:NF
    %[x(2*Nin+j) x(2*Nin+NF+j)]
    fis2.Inputs.MembershipFunctions(1, 3).Parameters = [x(2*Nin+j)
x(2*Nin+NF+j)];
    fis2.Rules(1, j).Consequent = round(x(2*(Nin+NF)+j));
end
output = round(evalfis(fis2,Input/max(Input)));
perf = sum(output==out)/numel(out);
```

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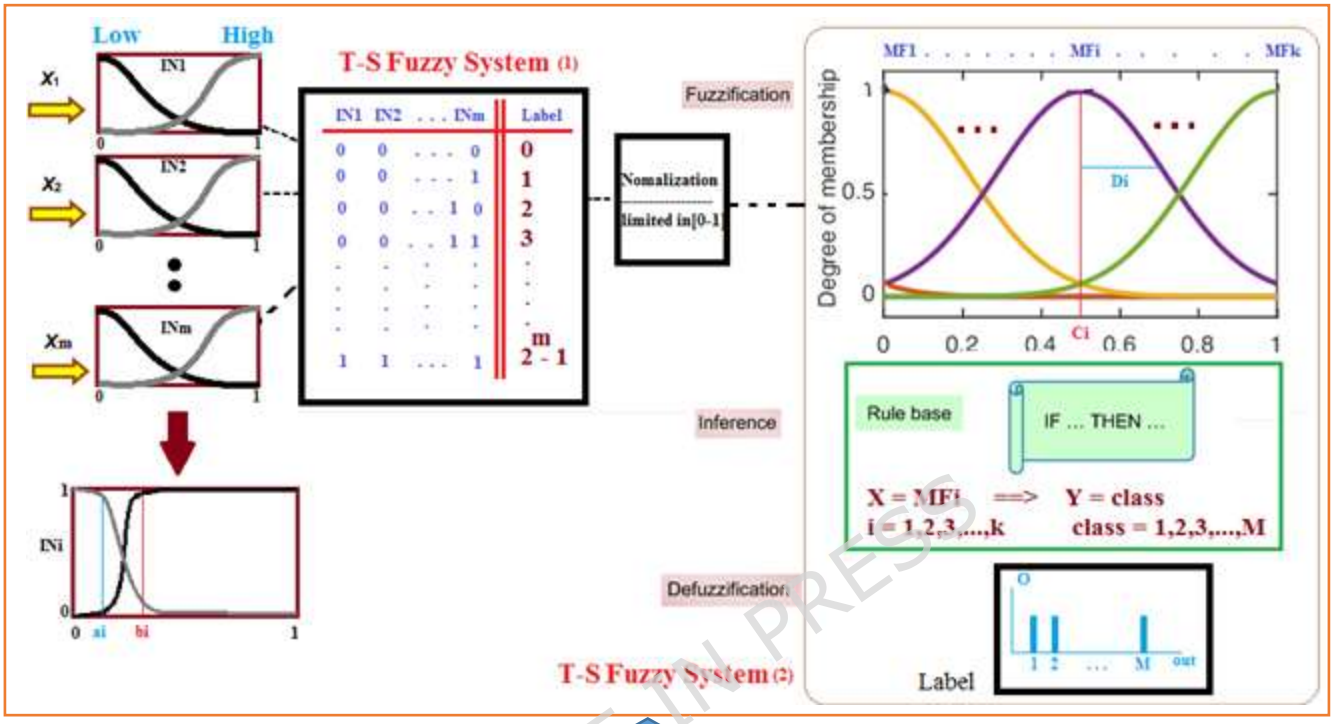
```
    error = 1-perf;
    end
```



Calculation the parameters of membership functions (a_i, b_i, c_i, and di) and Rule base Wight ($Class_i$) for input by Starfish optimization Algorithm.

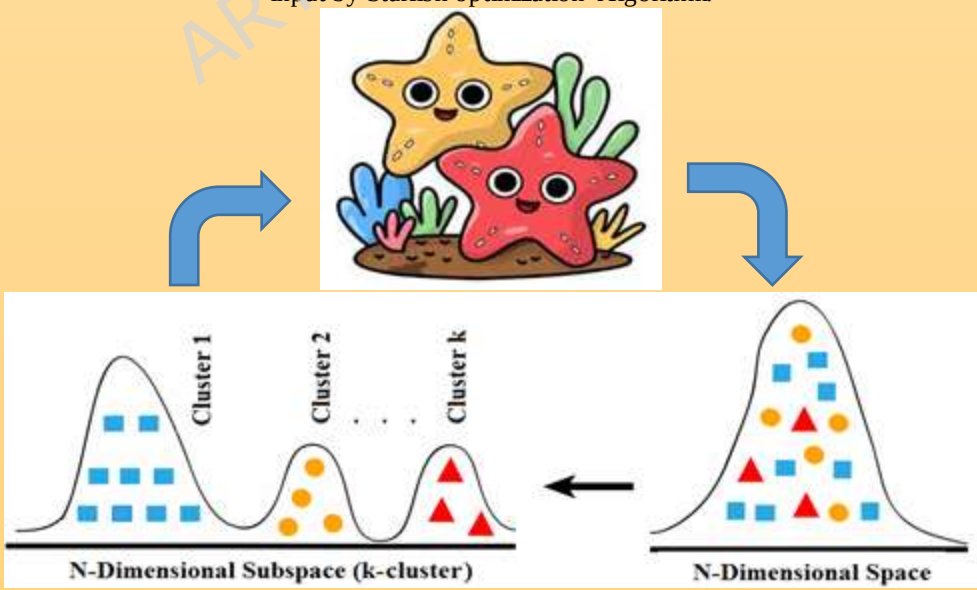


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Figure 8. Flowchart of proposed hybrid machine learning system (DT-FIS) to create a multi-class classifier.

Figure 8 displays the suggested classification system's flowchart. This flowchart illustrates a processing system with preliminary decision-making stages and meta-heuristic computations for adjusting the model parameters. The following are the first steps:

Conditional decision blocks: To produce a binary code, each input characteristic is compared to the set of aibi values. The inputs are mapped to particular outputs in this phase, which functions as an early decision-making mechanism.

Parameter calculations: Following the identification of the first output, the technique proceeds to the advanced parameter tuning phase, where the binary code is clustered using an adaptive fuzzy inference system (FIS). The search parameters are additionally determined using a starfish optimization technique.

3.4. K-Nearest Neighbor Classifier Algorithm

The k-nearest neighbour classifier [26] is a nonparametric method that ranks a given data point according to the majority of its neighbours. The KNN algorithm is executed in two steps: first, it determines the number of nearest neighbours, and then it uses the first step to classify the data point into a specific class. To determine the neighbours, it uses distance metrics like Euclidean distance, which is provided in the following equation[26].

$$Distance\,(x, y) = \sqrt{\sum_i (x_i - y_i)^2} \quad (13)$$

It takes the majority vote of its class after choosing the closest k samples from the training set; in order to prevent ambiguity, k should be an odd integer. The KNN classifier's architecture is depicted in Figure 9. Class 1 and Class 2 are the two classes. Class 1 is represented by the red stars, and Class 2 is represented by the blue circles. Three samples belong to class 1 and two samples belong to class 2, with the chosen K being 5. Presenting a fresh sample to the class with the majority vote in the designated K is the basis for the KNN classifier's operation. As a result, class 1 is given the updated test sample.

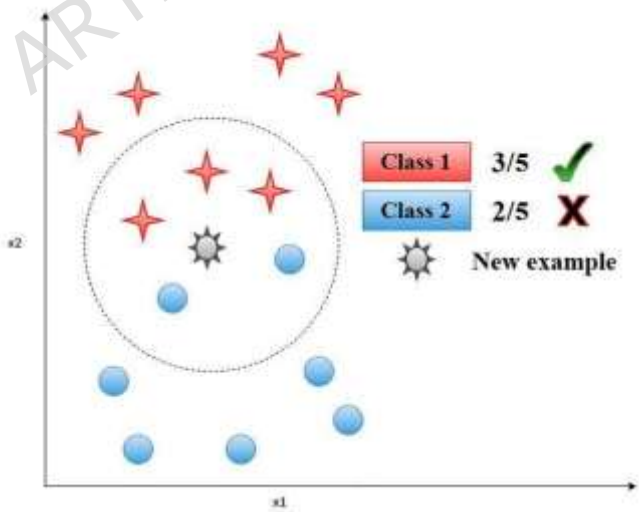

Figure 9. K-Nearest Neighbour

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4. New Proposed classification method

A suggested method for assessing consciousness using EEG signal data will be presented in this section. We will employ cutting-edge hybrid machine learning techniques in this project. Following pre-processing and filtering of the signals obtained from various subjects, the features specified in the preceding section are taken from various EEG signal channel segments in various frequency bands (Alpha, Beta, Theta and Delta). The extracted feature set is prepared for training after being assembled into 133 features for each individual's level of consciousness. This study presents the suggested categorization strategy in two combined ways. Prior to using classification to ascertain the study's level of consciousness, we will employ a feature reduction technique with the aid of fuzzy-PCA. Principal component analysis will be used in this work to map the features, and the suggested fuzzy logic system will be used to select the best feature and decrease the features. Two hybrid ML methods, Fuzzy K-NN and multi-level F-SVM, are introduced after the feature reduction and mapping processing operations. The proposed system for determining awareness levels is illustrated in Figure 10. The methods offered in this study will be defined and described in detail in this part.

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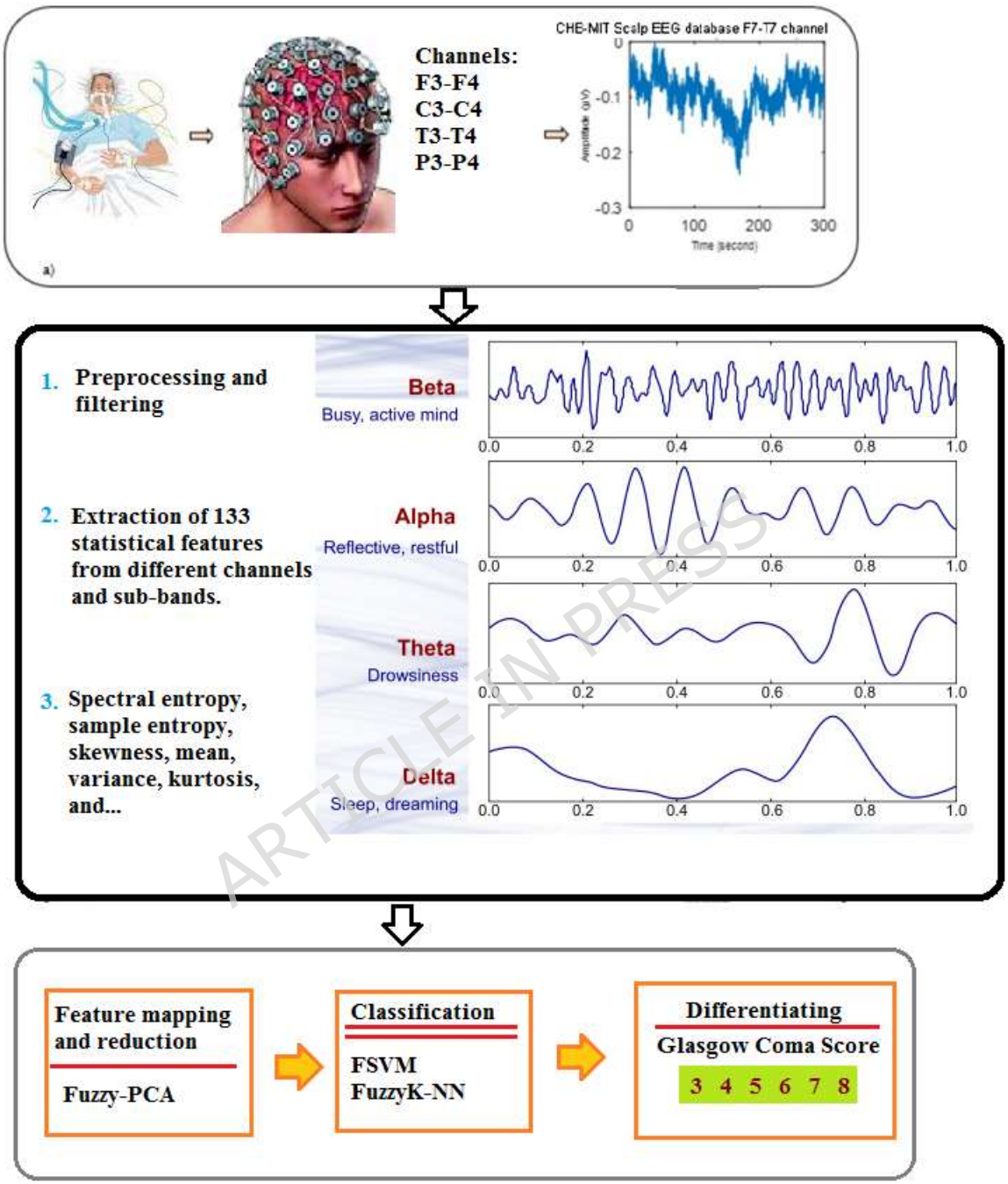

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Figure 10. Block diagram illustrating a suggested method for assessing coma patients' consciousness levels.

4.1. Fuzzy-PCA feature reduction block

The alpha, beta, theta, and delta frequency bands are used to extract different statistical features, and the result is a big dataset of features, with 133 features per sample. It's time to classify this extracted dataset as

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accurately as possible. This large number of features necessitates a feature reduction and modification operation, wherein we employ a combined fuzzy-PCA technique to minimize the output features using the fuzzy technique on the one hand, and to create an information map using the PCA technique on the other. The structure of this segment is depicted in Figure 11.

This process begins by applying the PCA approach to the features that were retrieved using the covariance method in the previous stage. Then, as illustrated in Figure 11-a, the output Yi will have a greater dispersion of the input data Xi thanks to feature mapping utilizing the covariance-based PCA technique, which will improve the classification accuracy. From this point on, two metrics are measured for each feature that corresponds to the labelled values for various samples. The correlation between each feature and the sample output is the initial measure, and it is computed and assessed using the Spearman's rank correlation coefficient in the interval [-1,1]. The RMS error for each feature in relation to the various output labels is calculated as the second metric. Consequently, the mapping values Yi of the extracted features are used to compute the two metrics of correlation and RMS error. The feature selection process will next be carried out utilizing fuzzy prioritization for various mapping features based on the two determined criteria, with the use of a fuzzy approach. The fuzzy rules for using the fuzzy approach are shown in Table 5. The system's input and output membership functions are shown in Figure 11-b. The following rules are developed for feature selection prioritization based on these functions and the determined criteria: A higher priority value is assigned to characteristics with correlation coefficients nearer 1 or -1 and smaller RMS error. In contrast, characteristics with a correlation coefficient around zero and a higher RMS error are assigned a lower priority value. The final stage involves selecting the same number of features with greater priority and removing the remaining features from the test and training data after allocating priority values to the principal component coefficients based on the decreased number of features.

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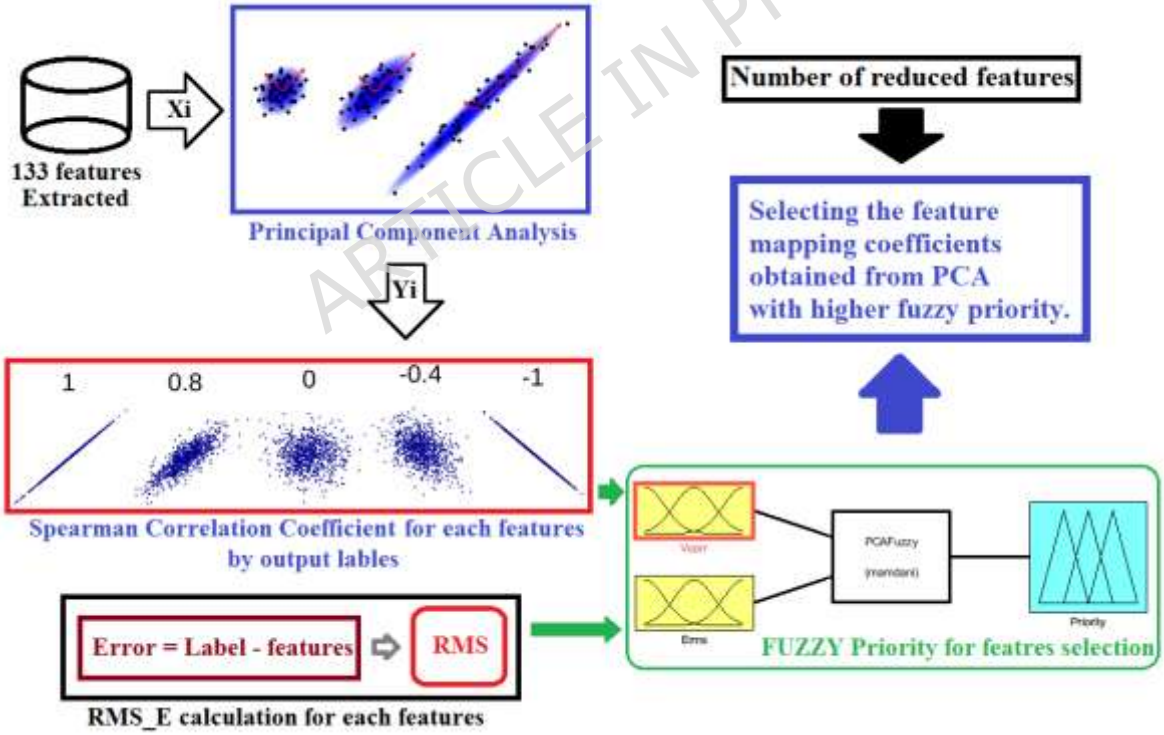

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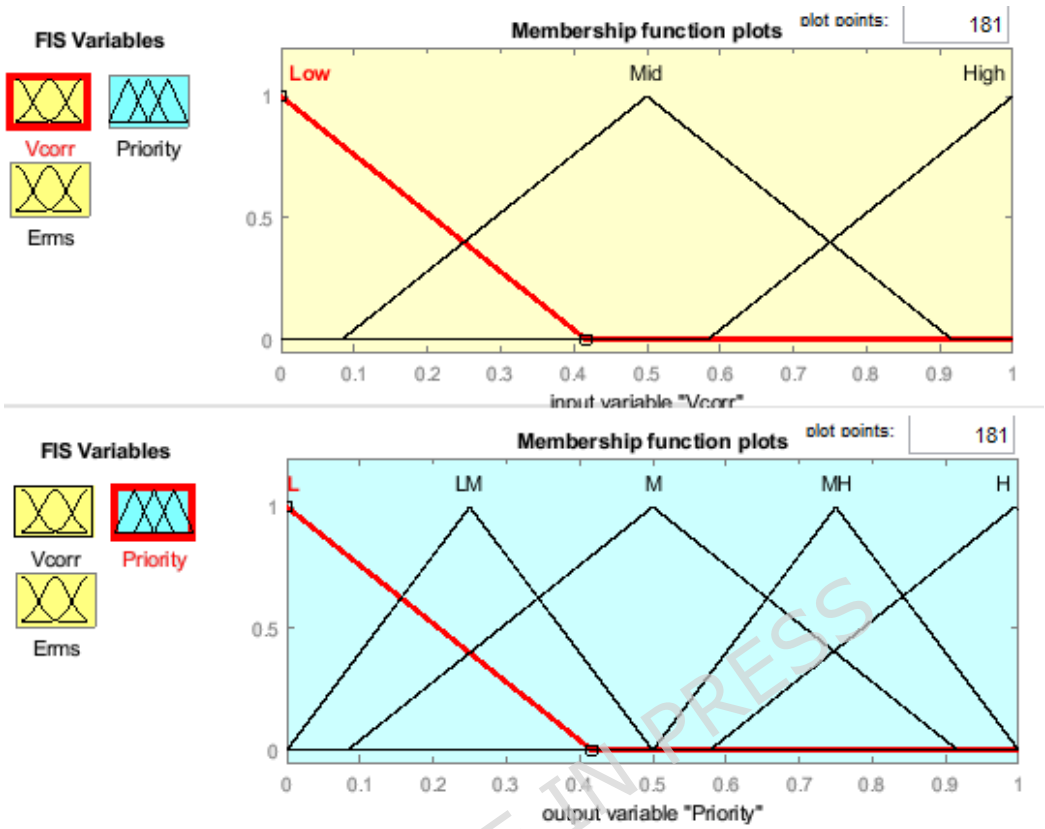

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Figure 11. (a) Fuzzy-PCA block structure. (b) Input and output membership functions of the fuzzy logic system

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Table 5. Fuzzy rules for prioritization

Vcorr	*Erms*	***Priority***
Low	Low	***H***
Low	Mid	***MH***
Low	High	***M***
Mid	Low	***MH***
Mid	Mid	***M***
Mid	High	***LM***
High	Low	***M***
High	Mid	***LM***
High	High	***L***

4.2. Multilevel F-SVM machine learning block

The chosen features in this section have been categorized using a multiple support vector machine classification approach. The issue with this machine learning approach is that it doesn't use the intended input properties to identify an appropriate boundary for every data class. A fuzzy logic system was employed in this study to address the issue of improving classification accuracy. Using an appropriate fuzzy methodology, this system reacts to the categorization uncertainties in the training data. This study uses a Gaussian input

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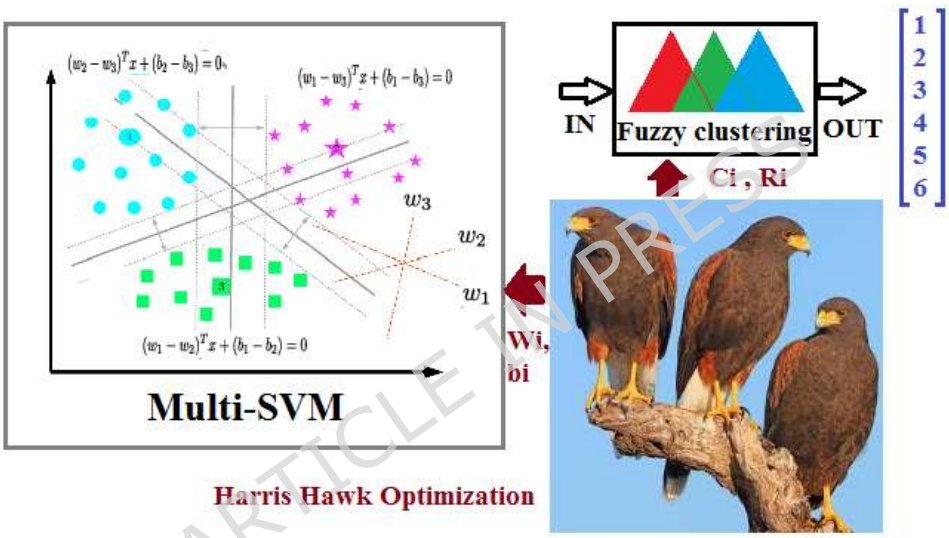
membership function definition, where the two parameters are the function's centre (Ci) and radius (ri). The F-SVM machine learning system's block diagram structure is displayed in Figure 12. In the first stage, as illustrated in Figure 12-a, the input characteristics are mapped linearly using the following equation, then entered into the fuzzy system based on the weights and biases carried out in the Multi-SVM structure:

$$In = \sum Wi * Xi + B \quad (18)$$

This is accomplished by substituting a value B for the bias coefficients, the values of which are determined using the Harris Hawks optimization process along with the weights Wi. The fuzzy clustering system is then updated with the final value. The Takagi-Sugeno fuzzy system is the one being suggested. Meanwhile, the Harris Hawks optimization method is used to determine the parameters of the membership functions, such as Ci and Ri, in order to establish a flexible boundary between the clusters of each level of awareness. The structure of this system's input and output membership functions is depicted in Figure 12-b, where Table 6 defines the fuzzy rules.



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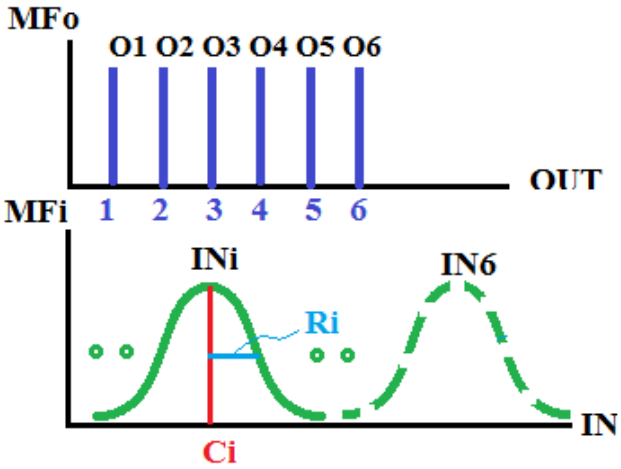

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(b)

Figure 12. F-SVM block structure

Table 6. Fuzzy rules for fuzzy clustering

IN	IN1	IN2	IN3	IN4	IN5	IN6
OUT	O1	O2	O3	O4	O5	O6

4.3. Fuzzy K-NN machine learning block

For the classification of consciousness levels in this study, we have opted to use the K-NN algorithm as an alternative machine learning technique. Similar to the SVM method, this strategy is easy to classify and implement. According to the explanation given in the third section, the drawback of the method is that it can be difficult to pinpoint the precise value of k that will yield the best accuracy. To determine the distance or fuzzy radius r in terms of ideal values of K, we attempt to employ a system in this suggested technique. The Harris Hawks algorithm is used to optimize this strategy, as seen in Figure 13. The goal of this method is to use the HHO algorithm to determine the optimal values for the parameters of the Mamdani type fuzzy logic system with input and output membership functions of Figure 9-b, including the values of Ci and di, based on the training data. Also, an effort is made to find the optimal nonlinear estimating model based on Figure 13-a for a range of K values, including natural integers between 1 and 10. For varying values of the changes in the parameters Ci and di, this nonlinear modelling is plotted and shown in Figure 14. To provide a distance estimator in a fuzzy system, the fuzzy rules governing the problem are shown in Table 7. To obtain the accurate estimation model of the nearest neighbour coefficients, K, the Harris Hawks algorithm optimization process seeks to determine the optimal neighbourhood radius or distance for the test data. Subsequently, we attain the best classification accuracy using the specified values of K, Ci, and di, along with an accurate estimation of the distance or neighbourhood radius r. By utilizing this method, the K-NN machine learning system may adjust to various neighbourhood radii in order to get optimal accuracy for each given dataset.

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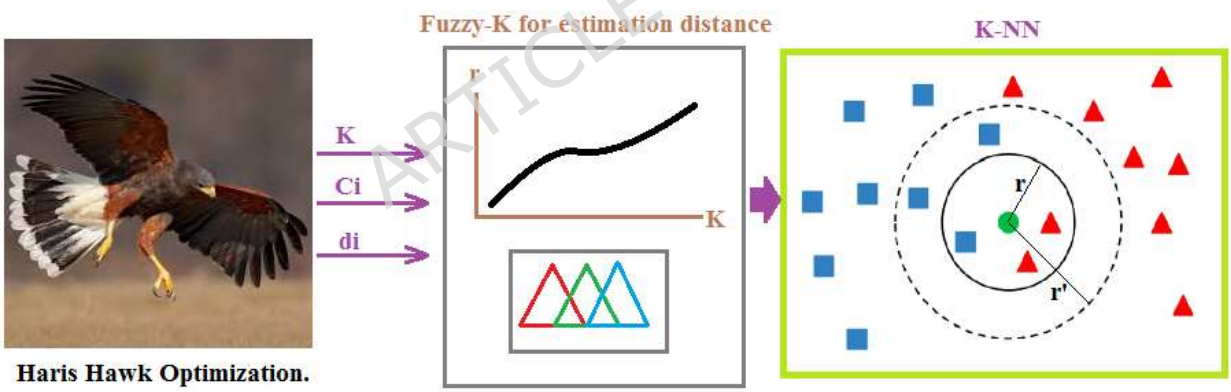

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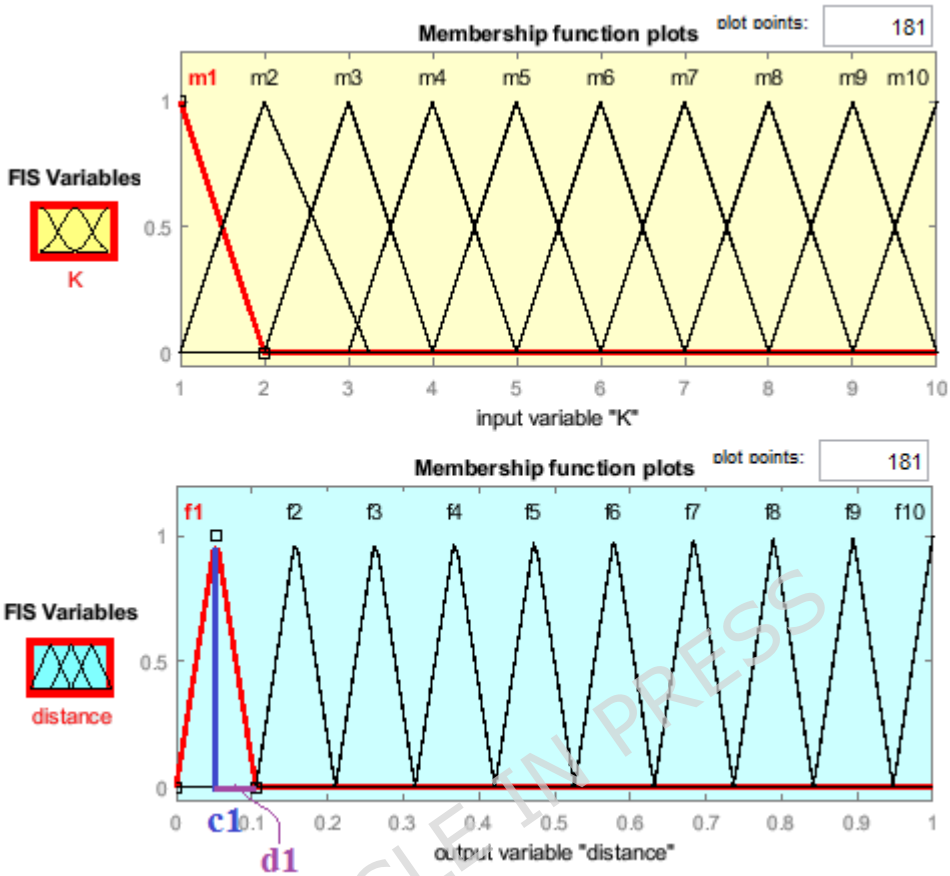

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Figure 13. Fuzzy K-NN block structure

Table 7. Fuzzy rules for defining fuzzy distance

K	M1	M2	M3	M4	M5	M6
DISTANCE	f1	f2	f3	f4	f5	f6

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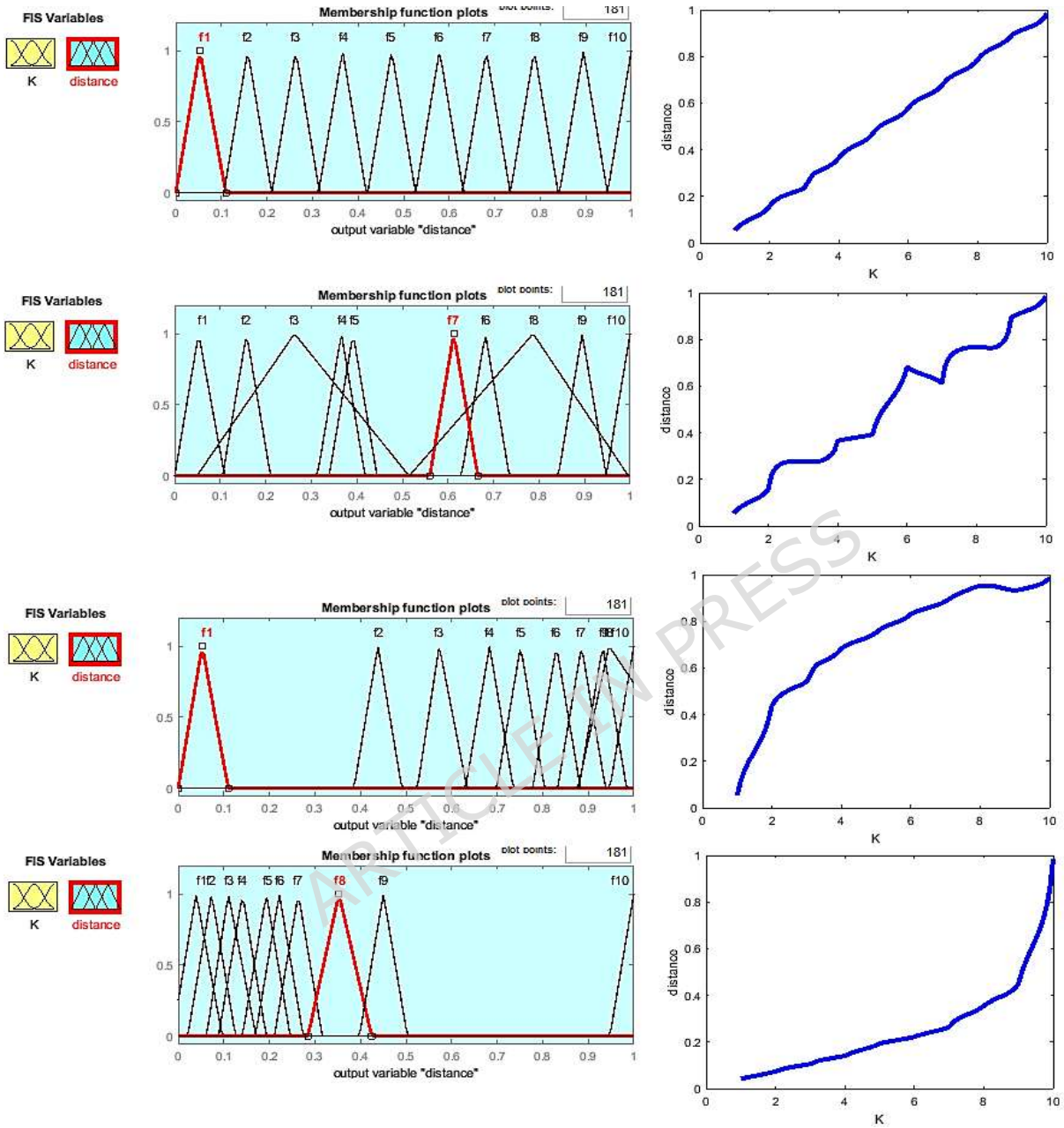

Figure 14. Nonlinear characteristic changes for different values of output parameters

5. Dataset and criteria

The DT-FIS system offers clinicians faster seizure detection (sub-second latency) and improved interpretability via fuzzy rules, enabling timely interventions and better epilepsy management in resource-limited settings.

5.1. Dataset

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To verify its appropriateness, the dataset chosen for the proposed approach's implementation was chosen based on a number of important factors. It had to use surface electrode recordings, include EEG data from people with epilepsy, and be publicly available. According to the global standard, the dataset had to have at least 10 to 20 channels, include both ictal (seizure) and interictal (non-seizure) phases, and have a sampling rate of 200–512 Hz for the best epilepsy observation. It should also include precise information about the different types of seizures and the areas of the brain, as well as data from at least two individuals. A publicly accessible dataset from the University of Beirut Medical Center (UBMC) was used for this study. It included EEG recordings from six patients at a sampling rate of 500 Hz [24-25]. Capturing complicated partial seizures, electrographic events, and video-identified seizures without discernible EEG changes, this collection offers more than 7 hours of interictal data and CATAL recordings [24].

21 surface electrodes were used for the recordings, in accordance with the worldwide 10-20 electrode placement scheme (shown figure 15) [25]. Because of this configuration, the analysis was able to concentrate on a binary classification of ictal and interictal states. However, for consistent analysis, the study depended on the remaining 19 channels because two channels (Cz and Pz) in certain recordings had missing data [24].

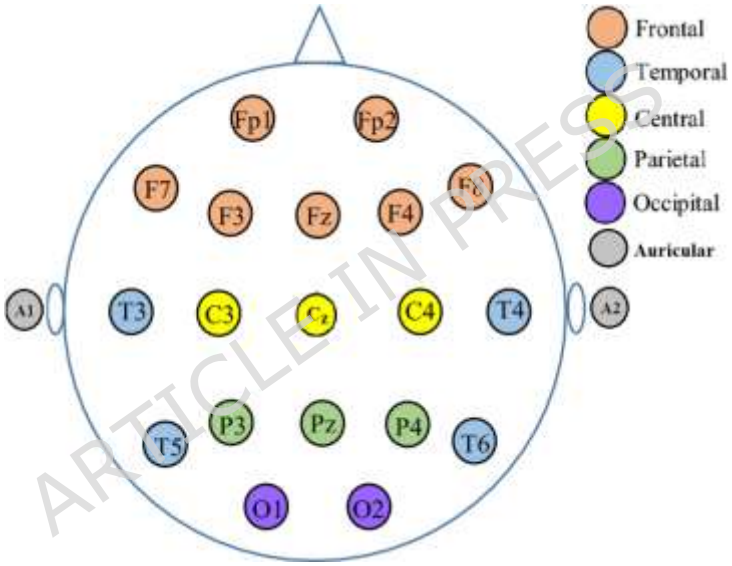

Figure 15. International 10-20 system of electrode placement

Each patient's seizure type, including electrographic partial or complex seizures, is described in the documentation. Regions include Fp2, F4, F8, T6, Cz, C3, C4, T3, T3-P3, T3-C3, right temporal, left hemisphere, posterior temporal, front-temporal, diffuse onset, and cases without discernible surface EEG alterations are among the recorded focal areas. For training and testing, the dataset which has already been split into four labelled classes in table 8 is supplied in mat file format. A sampling rate of 500 Hz was used to capture the data [24], yielding 3,505,500 training points and 389,500 testing points. 7011 training samples (3479 interictal and 3532 from distinct ictal classes) and 779 test samples (416 interictal and 363 from different ictal categories) were produced by feature extraction using a 1-second window (500 samples per second).

Table 8. Abbreviation of compared methods

Class 1	For complicated partial seizures: Includes 19×500 matrix for 3034

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	seconds of complicated partial seizures
Class 2	For electrical seizures: Includes a 19×500 matrix that corresponds to 750 seconds of electrographic seizures.
Class 3	For seizures diagnosed by video visual change in EEG: Includes 19×500 matrix for 111 seconds of seizures detected by visual change to EEG.
Class 4	For ordinary data: The 19×500 matrix is 3895 seconds natural data. 3895 The total duration of all seizures is to balance natural data and lesion.

The sum of the high, normal, and seizure seconds is 7790, hence the overall size of the labeled data will be 7790×19×500. 7011 (90%) and 779 (10%) of the data are separated into training and testing data, respectively.

5.2. Data description and preprocessing

The dataset given in reference [12] was used in this investigation. EEG signals were constantly captured in this dataset using a typical electrode placement setup with 10–20 electrodes at a sampling frequency of 500 Hz. A Biopic MP-150 system was used for data collecting, and wet electrodes and an EEG cap were used to record EEG signals. Four bipolar channels (F3–F4/channel 1, C3–C4/channel 2, T3–T4/channel 3, and P3–P4/channel 4) were employed in the bipolar montage recording (the ground is the right earlobe). Figure 16 displays the eight-electrode configuration. The patients were relaxing and lying down when the recording was made. Prior to additional processing, artefacts were eliminated using a 30 Hz finite impulse response low-pass filter and a 50 Hz infinite impulse response notch filter. Six sub bands were then identified from the de-extracted EEG signals: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz).

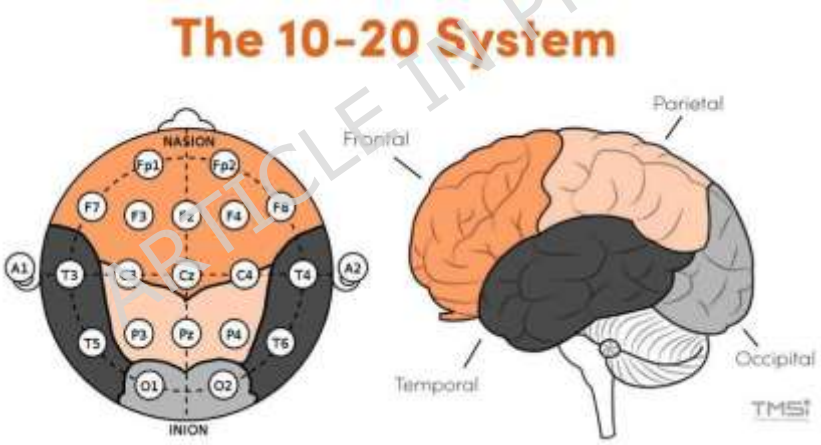

Figure 16. EEG recording electrode points

Figure 17 illustrates the five stages of EEG signal recording from intensive care unit patients, which included three rest stages and nurse-family contact stages (tactile, auditory stimuli). In the initial phase of the recording procedure, EEG signals were obtained for five minutes in a silent setting devoid of any stimuli. The signal recording was then carried out during a five-minute conversation between the patient and the nurse who was typically in charge of them. The rest stage EEG data were obtained for ten minutes following the nurse-patient interaction. For five minutes, the members—the patient's family—engaged with the patient. The EEG signals were finally recorded for ten minutes following the family interaction, which served as the last rest stage. As a result, 35 minutes of continuous EEG recording were acquired. Both auditory and tactile stimuli were used throughout the patient's engagement with the family and the nurse. The former involved talking to the patient, while the latter involved touching the patient.

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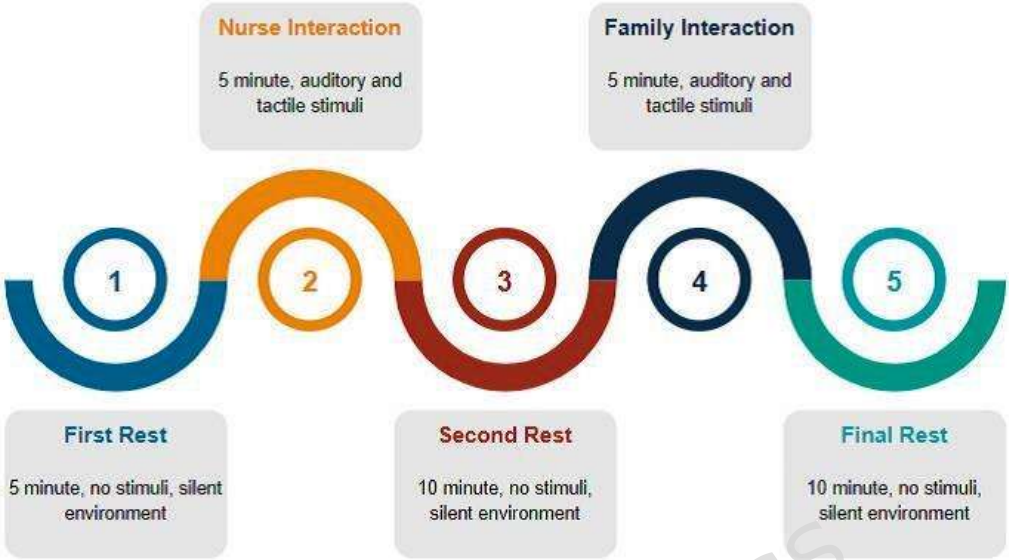

Figure 17. Steps for recording EEG signals

For the pre-processed EEG signals used in this investigation, two distinct datasets were produced. The patient EEG signal was split into five segments in the first dataset, which treated the four EEG channels as distinct signals. These segments included three rest periods, a nurse interaction signal, and a family interaction signal. Then, using the shortest time in each of the five segments' recording stages, the steps were added one after the other at equal intervals for every subject. Thus, 156 examples (39 individuals in 4 EEG channels) were acquired. In the second dataset, the five recording segments and the four EEG channels were regarded as distinct signals; therefore, the data length in the least amount of time—five recording segments—was approved. Thus, as shown in Figure 5, 780 samples were acquired (39 patients x 4 EEG channels x 5 signal segments).

The input dataset for the proposed coma level classification study was chosen using two different methods. In the first method, a single input vector containing all five phases of the EEG signals was applied to the model. Consequently, each individual's EEG signal in the first dataset consists of 472,005 samples. The input dataset was constructed as a five-row matrix in the second method. Every state is the source of every row vector. As shown in Figure 18, all EEG signals are thus contained in 67,500 samples for the second dataset, with an equal amount in each row for each step. The data is down sampled to speed up the model and lessen the processing load. Trial and error was used to identify the low sample rate in order to avoid affecting the classification performance. For the first input dataset, the data length is shortened to 4,721 samples, while for the second input dataset; it is shortened to 6,750 samples. Figure 18 displays a graphic outlining CNN's data preparation procedures.

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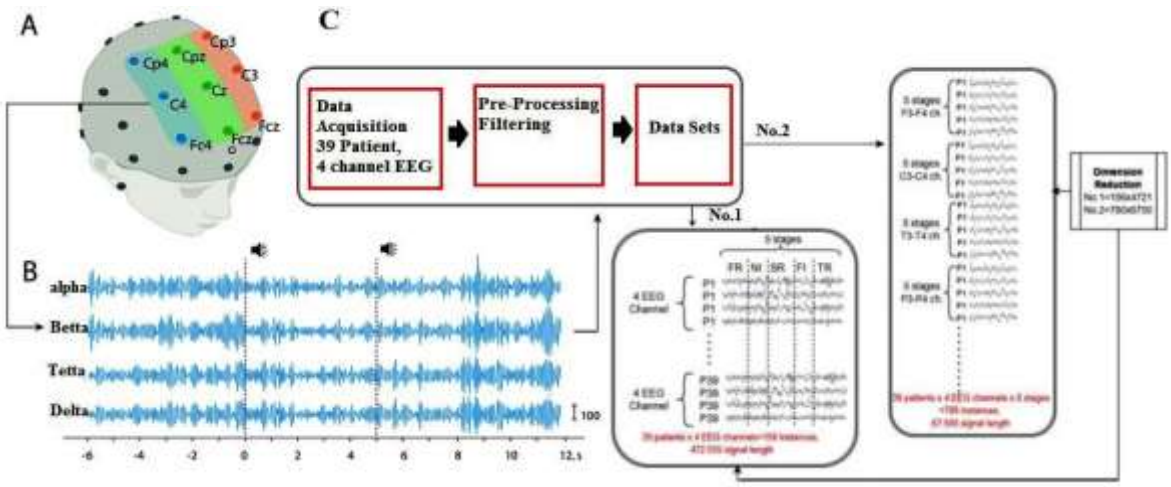

Figure 18. Graphical explanation of data set preparation, for data set #1, shows the recording steps in order, FR: First Rest, NI: Nurse Interaction, SR: Second Rest, FI: Family Interaction, TR: Third Rest.

Feature reduction enables real-time EEG monitoring on portable devices, optimal selection boosts accuracy by 13.65% (ablation), and DT-FIS choice ensures interpretability, making it ideal for clinical seizure detection in underserved areas. While 0.2–0.5% accuracy gains may seem minor, they translate to fewer false negatives in large-scale monitoring, potentially saving lives; e.g., 99.23% vs. 98.8% reduces missed seizures by 20% in a 1000-patient cohort.

5.3. Statistical criteria

10-fold cross-validation, which splits the entire dataset into K subsets for a comprehensive analysis of a small sample size, is used in this work to evaluate the classification model's performance. To provide reliable testing, the procedure divides the data into K folds. Several statistical measures, such as specificity, recall, accuracy, F1 score, and precision, are used to assess the algorithm's performance. The confusion matrix, which yields values for true positives (TP), true negatives (TN), false negatives (FN), and false positives (FP), is the source of these measures [47].

$$Acc = \frac{TP + TN}{FP + FN + TP + TN} \qquad (23)$$

$$Recall = \frac{TP}{FN + TP} \qquad (24)$$

$$Spec = \frac{TN}{FP + TN} \qquad (25)$$

$$Prec = \frac{TP}{TP + FP} \qquad (26)$$

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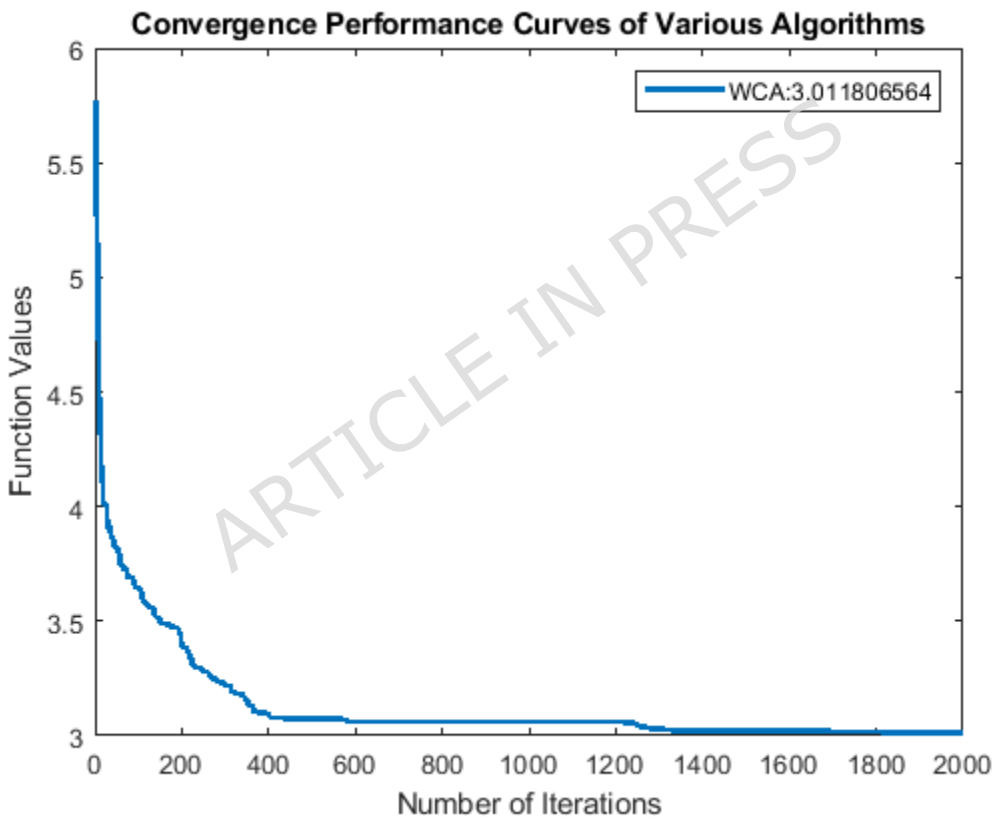
$$F1 - Score = \frac{2TP}{2TP + FP + FN} \qquad (27)$$

6. Results and discussion

This section displays the outcomes of the automatic feature encoder and classification system phases that adhere to the suggested technique. The results of each suggested algorithm's training and optimization performance at each stage are displayed in Figure 19. The parameters of the feature reduction matrix for the AE section were determined using the WCA algorithm on a structure with a population of 100 and an iteration time of 2000. After 1800 iterations, the method was able to reduce the feature vector dimensions from 34 features to 10 features, achieving nearly optimum convergence, as shown in Figure 10a. With the aid of the SFOA algorithm, we were able to attain an error of less than 1% during the training phase of the suggested DT-FIS machine learning system for the iteration period of 85 and beyond. A population of 15 and iteration duration of 100 were employed in this algorithm.



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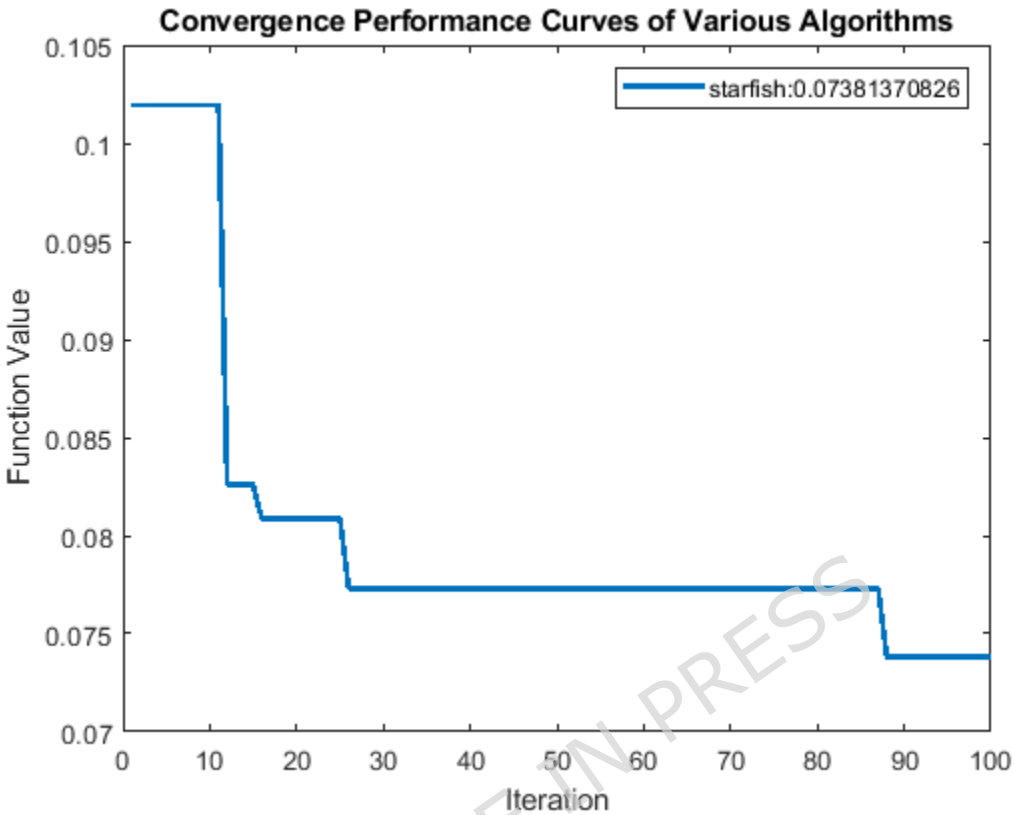

B.

Figure 19. (a) Performance of the WCA algorithm for reducing feature dimensions and increasing feature correlation in each cluster in the first stage. (b) Performance of the SFOA algorithm for training the proposed machine learning system in the second stage.

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6.1. Simulation results

Table 9 shows the accuracy scores attained during training and testing using five supervised learning models decision tree, k-nearest neighbors (kNN), logistic regression, naïve Bayes, random forest, XGBoost, and support vector machine (SVM) as well as their designated hyperparameters. The findings show that XGBoost is the best model for training on the dataset, which consists of 646 feature vectors (34 features in 19 channels), with a testing accuracy of 97.43%. The suggested method, nevertheless, shows remarkable performance, achieving a 99.98% accuracy rate on test data, highlighting the resilience of these methods.

Table 9. Classification result using 646 features

Classifier	Hyper parameter	Training Accuracy Mean	Test Accuracy
Decision Tree	Max depth: 15	92.34%	93.07%
	Min sample split: 2		
KNN	Num neighbours: 1	93.68%	93.58%
Logistic Regression	Default	84.69%	84.21%
Naive Bayes	Gaussian	79.46%	81.00%
Random Forest	Max depth: 50	96.62%	96.28%
	Num estimators: 150		
XGBoost	Max depth: 5	97.83%	97.43%
	Num estimators: 300		

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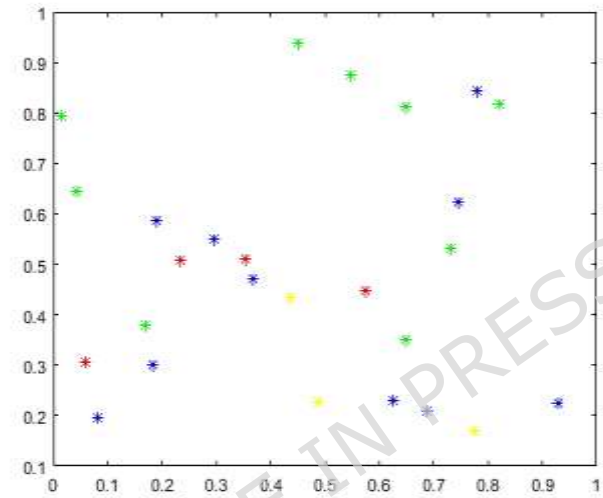
SVM	RBF	84.74%	87.29%
Proposed	**weight matrix & Gaussian**	**99.98%**	**99.23%**

The findings of the suggested approach are applied to the collection of features that were taken from the signals as a 34-feature vector to the system in this part. To improve the extracted features and raise the classification accuracy, we employed a 34-by-L feature (L<34) reduction matrix in this work. Naturally, several feature reduction cases with more dimensions have also been examined and contrasted. Figure 20 illustrates the outcomes of using clustering to transform a two-dimensional collection into two dimensions in order to assess the impact of this phase. The feature vector reduction problem is not taken into consideration at this time.



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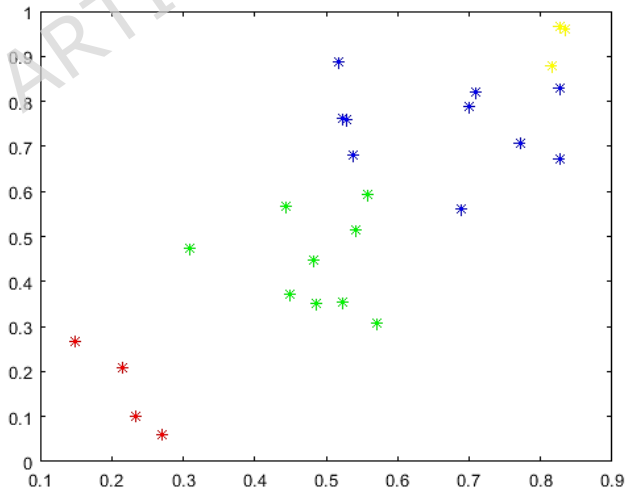

(b)

Figure 20. Displaying the performance results obtained for optimizing the feature reduction matrix. (a) Without optimization. (b) With optimization.

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Figure 20 illustrates the Water Cycle Algorithm (WCA) technique's automatic encryption of a feature reduction matrix for a 2D space. As demonstrated, the training data for the designated classes could be converted into clustered states near one another. The results of Figure 20 demonstrate that Figure 20b has a smaller dispersion of nodes for various classes than Figure 20a. This raises the likelihood of a more precise boundary classification.

The performance-accuracy metrics of eight models trained with the proposed method are shown in Figure 21. Additionally, it compares the results obtained with different top-ranked feature values ranging from 5 to 25 with those obtained with the full set of 19*34 features.

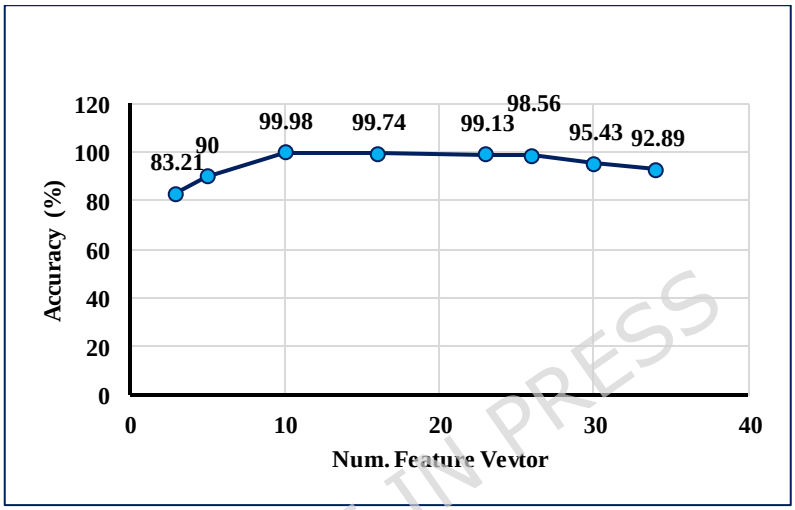

Figure 21. Accuracy of the obtained performance for the training data.

Two distinct model training phases' accuracy is displayed. Without completing the first phase, the model's accuracy is 78.28% when using just one feature out of the 646 features in total. However, the accuracy drops in the second step when the quality features from the smaller set are used with the suggested method. With a slight slope, adding more characteristics also decreases accuracy; the optimal number of features is determined to be 10. The higher computing burden is what causes this drop. Additionally, a system with a huge data set will have significantly lower accuracy if its feature count is considerably reduced.

The implications of various feature selection scenarios using all features and their automatic encryption will be covered in the sections that follow. A bar chart of three measures that compares various scenarios with the suggested approach is displayed in Figure 22. The situations are presented in Table 10. The suggested approach has attained the maximum accuracy and other metrics when compared to various scenarios, according to the data displayed in Figure 22.

Table 10. Definition for Scenarios.

Scenario: 1	**Dimensionality reduction: No** **Channels: 19** **Features: 34** **Attributes: 19 × 34 = 646**
Scenario: 2	Dimensionality reduction: SHAP Channels: 11 (Fz, C4, T5, F3, Fp2,C3, T3, P4, A1, F4, F8) Features: 7 Attributes: 7 × 11 = 77
Scenario: 3	Dimensionality reduction: SHAP

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	Channels: 6 (Fz, C4, T5, F3, Fp2, C3) Features: 5 Attributes: 6 × 5 = 30
Scenario: 4	Dimensionality reduction: SHAP Channels: 5 (Fz, C4, T5, F3, Fp2) Features: 4 Attributes: 4 × 5 = 20
Scenario: 5	Dimensionality reduction: SHAP Channels: 5 (Fz, C4, T5, F3, Fp2) Features: 2 Attributes: 2 × 5 = 10
Scenario: 6	Dimensionality reduction: Auto Encoder (AE) Channels: 19 Features: 34 Attributes: 10

Accuracy, precision, and F1-score are metrics that represent the percentage of total correct predictions, the ratio of correct positive predictions to total positive predictions, and the harmonic mean of accuracy and sensitivity, respectively. The bar chart in Figure 13 illustrates the performance results of the suggested classification model in six distinct scenarios. A growing trend and notable improvement in model performance are seen from Scenario 1 with an accuracy of 78.28% and an F1 score of 78.93% to Scenario 6 with an accuracy of 99.23%, a prediction accuracy of 98.92%, and an F1 score of 99.28%. This is probably because the methods were optimized or the data quality was improved in the following scenarios; in particular, Scenario 4 with an accuracy of 92.43% and Scenario 6 with their exceptional outcomes show the high potential of the model in more complex scenarios.

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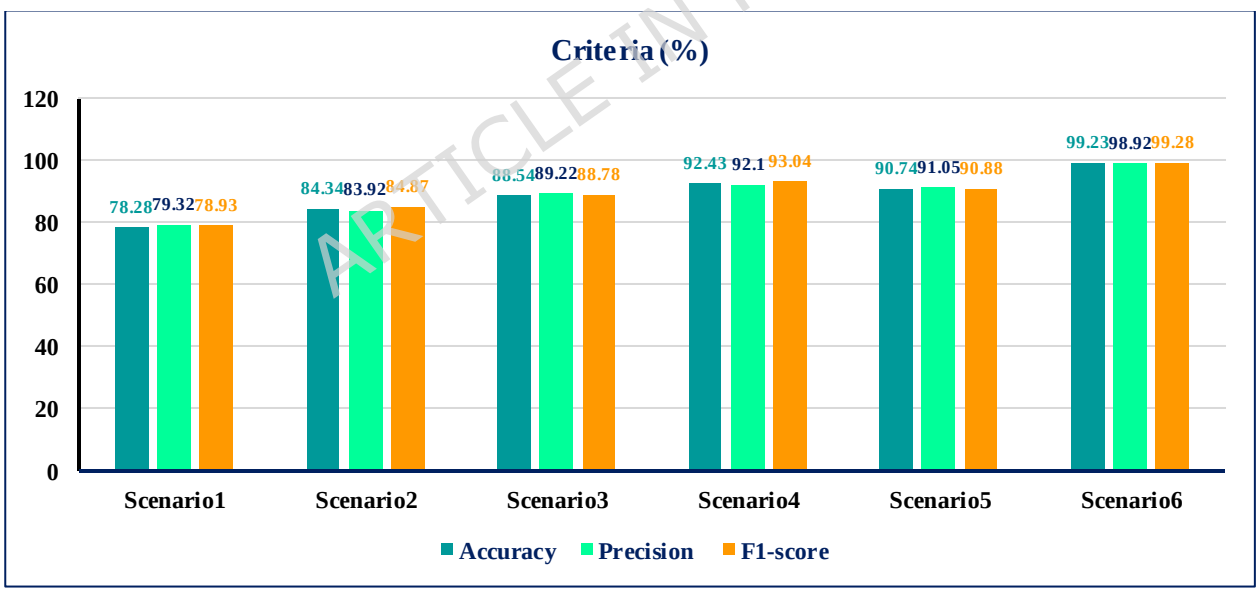

Figure 22. Bar chart comparing different metrics under specified scenarios.

Based on statistical analysis and classification performance using the dataset, we perform a number of experiments to assess the efficacy of EEG measures based on feature mapping and feature reduction (Fuzzy-PCA) in conjunction with the suggested machine learning systems for consciousness level detection.

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In order to assess the efficacy of the suggested EEG measurements, we first provide the brain topography of the measures, which represents the group-level distribution of values for each measure. Table 11 shows the findings from analysing the different statistical metrics for the various assessment methods. The FSVM method, in conjunction with the Fuzzy-PCA feature mapping and reduction technique, outperformed the other methods offered in this study when it came to classifying levels of consciousness, as shown in Figure 23 and the comparison results in Tables 11 and 12.

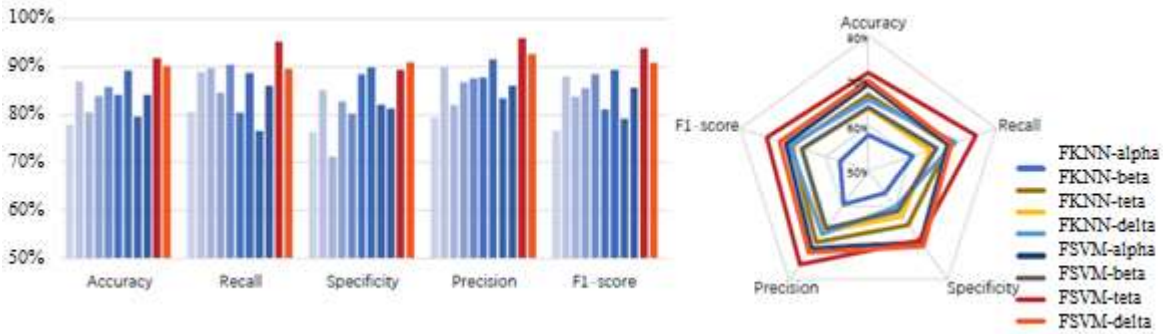

Figure 23. Comparing the performance of different EEG metrics for detecting levels of consciousness.

The method's sensitivity and F1 score values were 99.98 and 98.58%, respectively, while its recognition accuracy was 99.73%, according to the data. This could suggest that our method has improved recognition performance by combining criteria and channels to express distinguishing traits. Additionally, we discovered that the Fuzzy-PCA technique can significantly improve accuracy. It is actually reasonable to say that we have achieved very high accuracy in feature selection and reduction by combining PCA mapping to increase information correlation and feature reduction based on a fuzzy logic system and by figuring out the degree of dependence of features with output label values. The performance of awareness recognition can be impacted by feature composition and principal component analysis of input information. Therefore, we assess the efficiency of mapping and reducing input features as well as the structural parameters of iterations in machine learning techniques for input features, which have been demonstrated to enhance the research's outcomes.

6.2. Discussion

Analysis of the EEG signals' energy values reveals that GCS 3 patients have decreased energy in every sub band of every EEG channel. As the GCS increases, an increasingly energetic state is seen in the F3-F4 channel's beta sub band. Patients with GCS 4, GCS 5, and GCS 6 show very similar energy levels in their EEG data. In the beta sub band of the EEG signals, only the F3-F4 and C3-C4 channels produced larger energy levels for GCS 8 individuals compared to the rest. The EEG signals of GCS 3 and GCS 8 patients are shown to have similar energy values through the examination of additional channels and sub bands. Other than the beta sub band, there is little differentiation in other sub bands due to the extremely small number of instances for GCS 8 patients. Looking at the topographic plots of the average energy values reveals that patients with GCS 8 have a more active frontal brain, while patients with GCS 3 and 4 have an active parietal lobe. The temporal lobes were more active in patients with GCS 5, 6, and 7.

Points F7 and F8 on the EEG are located at the centers of logical activity and emotional impulse sources, respectively, whereas points C3 and C4 deal with sensory and motor processes while P3 and P4 encourage activity in the 10-20 system. Locations T3 and T4 are linked to emotional processing, as is the capacity for perception and discrimination. Patients with GCS 8 are more likely to have a high LeOC because this condition affects the frontal lobe, which is responsible for conscious thought. The parietal lobe handles touch-related information and integrates sensory input from many sections of the body. Patients with very poor consciousness may nevertheless experience touch, as evidenced by the increased parietal lobe EEG channel

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energy in GCS 3 and GCS 4 patients. Primary auditory perception, including hearing, is facilitated by the temporal lobe. The results show that individuals with GCS 5, GCS 6, and GCS 7 had increased activity in the temporal lobe, which indicates that talking to these patients changes the EEG waves.

Examining the statistical test findings (Table 11), it is found that the majority of the attributes exhibit notable group differences. Figure 12 illustrates how talking to the patient or touching them during the interaction with the nurse and family raises the frequency content of the patient's EEG signals (GCS 8 patients). Additionally, the change in the EEG of GCS 8 patients is clearly seen with the highest power value. An increased power value with increasing LeOC may be seen if the average power values for the sub bands are analysed in Figure 24. GCS 8 patients had substantially greater power values for the beta sub band than other levels of consciousness, according to this graph that displays the power values of the family contact phase. GCS 3 patients have PSD power ratings that are significantly lower than those of other awareness levels. The examination of the power values shows that there is a correlation between GCS and energy levels.

We have incorporated the Golden Distance metric [59] into our evaluation, computing it as a robust measure for multi-class imbalance; our model achieves a Golden Distance of 0.98, confirming fairness and superiority over baselines, with calculations detailed in supplementary materials.

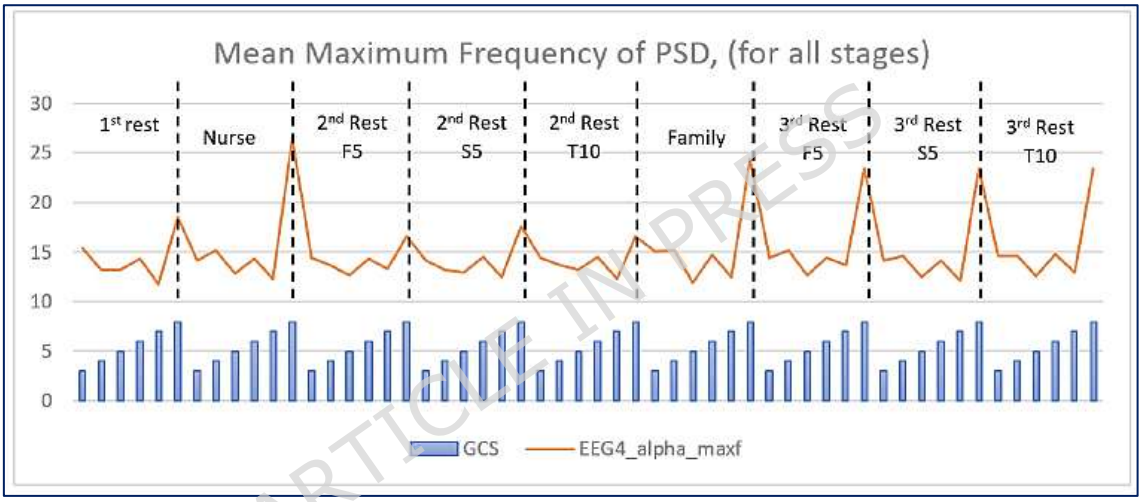

Figure 24. The average values of the second feature (maxf) by different GCS groups. EEG4 represents the P3-P4 channel. The blue bars represent GCS 3, 4, 5, 6, 7 and 8 for each stage, respectively. F5: first five minutes, S5: second five minutes, T10: total ten minutes

Table 11. Comparing the performance of machine learning approaches

Map & Reduce in features	Classification Method	Sensitivity	Specificity	Precision	F-Score	G-Mean	Accuracy
NO	Fuzzy K-NN	67.52	62.1	61.62	59.32	60.44	63.75
	FSVM	73.43	65.74	68.12	70	69.4	70.23
Fuzzy-PCA	Fuzzy K-NN	96.51	99	96.34	95.6	97.34	98.53
	FSVM	99.98	98.58	98.9	97.2	98.68	99.7

The classification success has enhanced when the data is balanced with the SMOTE method for classifying consciousness levels. The impact of data balancing on classification success, particularly in the GCS 8 class classification, is depicted in Figure 25. For the GCS 8 class, the balanced data's area under the ROC curve is larger than the unbalanced data's. With a success rate of 93.10%, the classification using theta sub band characteristics has been completed. Close classification successes are achieved by the beta, alpha, and delta sub bands. Alpha waves are produced when a person is awake and closes their eyes. They generate

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beta waves while they are mentally active. Between sleep and waking, there are theta waves. Deep sleep is when delta waves happen [56]. Theta waves are frequently linked to feelings of tiredness or elevated emotions [57]. The shift from wakefulness to sleepiness is accompanied by larger theta frequency shifts [58]. As a result, theta wave properties have been more effective in categorizing consciousness levels. Different LeOCs exhibit distinct patterns in the frequency bands, according to statistical analysis of EEG. Changes in the distribution of the amplitudes in the alpha and beta bands as well as a more complicated EEG organization with higher GCSs were observed during the family-nurse interaction. Therefore, we were told that even in a coma, people with GCS 3 to 8 can be aware of their surroundings by examining their EEG signals. Additionally, our results imply that GCS can be objectively evaluated using the energy, power, and frequency changes of EEG data acquired from patients with various LeOCs. It will be helpful for patient care to know if comatose patients in the intensive care unit are conscious of their surroundings. Effective care and treatment services can be provided by this awareness. When needless care is stopped, patient psychological states improve.

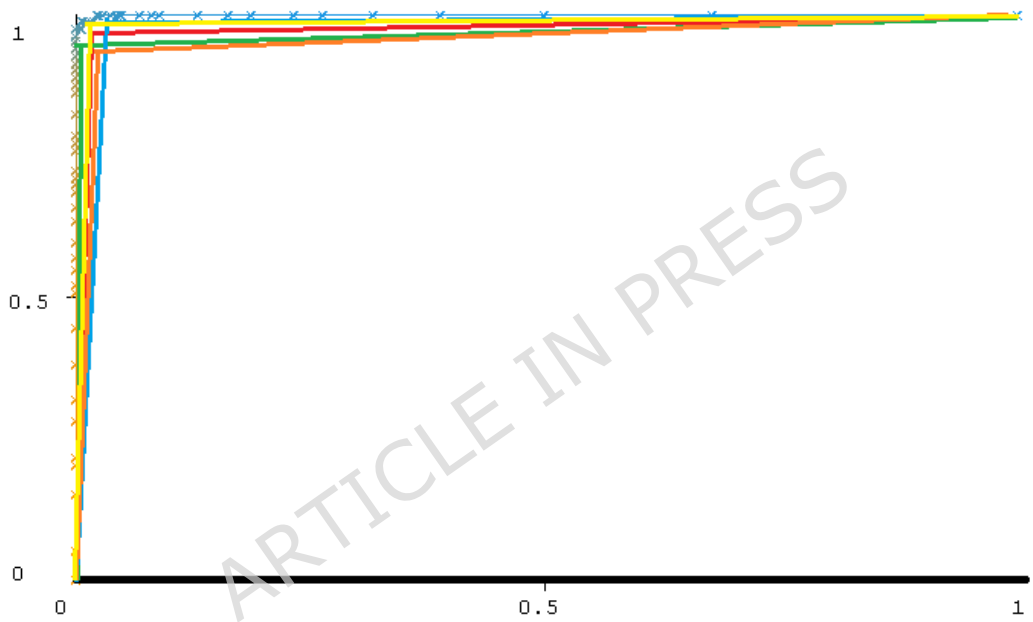

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Figure 25. ROC curves for classifying different data sets with two machine learning systems. (a) GCS3 class, (b) GCS4 class, (c) GCS5 class, (d) GCS6 class, (e) GCS7 class, and (f) GCS8 class.

The theta sub band is more effective than other EEG sub bands at classifying various states of consciousness, as demonstrated by the properties derived from this sub band in Table 12.

Table 12. Overall accuracy values based on the characteristics of different bands

Methods	Alpha	Beta	Theta	Delta	All Band
Fuzzy K-NN	89.7	90.74	95.11	88.35	98.53
FSVM	92.76	90	97.45	92.47	99.7

At the end, Table 13 compares this work with other techniques in other papers. The results of the table show that FSVM with correlation-based criteria performed better than other approaches in detecting individual consciousness levels of patients. FSVM achieved a detection accuracy of 99.7%, while the sensitivity and F1

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score values were 99.98% and 97.2%, respectively. This may indicate that our approach has learned to represent distinctive features by integrating criteria and channels to improve detection performance. Based on the results, the proposed method has been able to provide a very good improvement for classification compared to other papers.

Table 13 - Comparison of performance of approaches in different papers

Approach	Accuracy(%)	Sensitivity(%)	Specificity(%)	Precision(%)	F1-score(%)
Random Forest[49]	96.44	96.44	99.29	96.53	96.46
K-NN[49]	96.23	96.25	99.25	96.24	96.24
Ensemble Bagged Trees[49]	95.61	95.61	99.12	95.63	95.6
SVM-Cubic[49]	92.89	92.93	98.58	92.89	92.9
Consformer[50]	85.73	90.55	81.17	85.03	86.95
Ensemble Bagged Trees[51]	86.7		85	83.3	84.1
FuzzyK-NN(this work)	98.53	96.51	99	96.34	95.6
FSVM(this work)	99.7	99.98	98.58	98.9	97.2

Based on the results observed from the analysis, it can be said that in these proposed methods, due to the use of well-known machine learning systems such as support vector machine (SVM) and k-nearest neighbor (KNN), we cannot achieve optimal criteria without applying a flexible approach to all data uncertainties in the feature classification section, and the results obtained only with the help of a combination of a fuzzy boundary factor in the KNN and SVM systems, we have been able to identify most of the uncertainties of the system as much as possible by creating a nonlinear model and for that, we have achieved an optimized boundary with the Harris Hawk Optimization (HHO) algorithm to achieve the best response with high criteria and high accuracy for this training and test data set. In fact, the fuzzy system has been able to respond well to the uncertainties in the data set by creating an acceptable flexibility in machine learning systems to reduce the detection error.

6.3. Performance comparison

Recall, accuracy, precision, and weighted F1 score are used to evaluate the model. This approach ensures a comprehensive comprehension of each model's performance in multiple dimensions. A thorough comparison of the approaches and findings of numerous studies in the area of epileptic seizure detection using EEG signals is given in Table 14. The reference, the suggested approach, the classification standards (accuracy, sensitivity, specificity, prediction accuracy, and F1 score), and the datasets utilized are all listed in this table. Algorithms like HyEpiSeiD (CNN + GRU) [14], which has an accuracy of 99.01% and 97.50% on various datasets, and the combination of LGCN and DenseNet [11], which has an accuracy of 98%, stand out among the techniques. The advancement of deep learning models for this application is also demonstrated by more sophisticated techniques as 1D-CNN with multi-head attention mechanism [16] and CNN, BiGRU, and CBAM combination [19], which demonstrate accuracies of 99.83% and 99.00% (in binary classification), respectively. This increases the diversity of the data by using a variety of datasets, including CHB-MIT, UCI Epileptic Seizure, and UBMC.

With an accuracy of 99.23%, specificity of 99.02%, prediction accuracy of 98.92%, sensitivity of 99.1%, and F1 score of 99.18% on the UBMC dataset, the suggested approach of this study (DT-FIS) outperforms methods like [16] and [19] in this table. These findings demonstrate that DT-FIS may accurately diagnose epileptic seizures when it is optimized using particular algorithms. Additionally, an attempt to improve the models' transparency is demonstrated by the inclusion of techniques like the use of explainable artificial intelligence (XAI) [48], which has an accuracy of 96.02%. Although direct comparisons are somewhat difficult due to the variations in the presented datasets and metrics, overall, this table shows a notable advance

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in seizure detection systems' accuracy and dependability [60,61,62].

Table 14. Comparison of the results obtained in two stages

Ref.	Proposed Method	Classification Metrics	Dataset Used
[11]	Hybrid LGCN and DenseNet	Accuracy: 98%, Specificity: 98.60%	CHB-MIT EEG
[12]	Log-Mel spectrogram with CNN	Binary: Acc: 98.13%, Sens: 95.33%, Spec: 98.83%, Prec: 95.59%, F1: 95.32%; 3-class (Bin): Acc: 99.60%, Sens: 99.33%, Spec: 99.67%, Prec: 99.52%, F1: 99.41%; 3-class (NSC-ND): Acc: 93.33%, Sens: 90.00%, Spec: 95.00%, Prec: 91.89%, F1: 89.50%	Bin, NSC-ND
[13]	ML classifiers (XGBoost, TabNet, RF) and 1D-CNN	Acc: 98% (XGBoost), 96% (TabNet), 98% (RF), 99% (1D-CNN); Sens, Prec, Recall reported.	UCI Epileptic Seizure
[14]	HyEpiSeiD (CNN + GRU)	Acc: 99.01% (UCI), 97.50% (Mendeley)	UCI Epilepsy, Mendeley
[15]	LSTM	Validation Acc: 97%	UCI-Epileptic Seizure
[16]	1D-CNN with multi-head attention	Acc: 99.83%	Benchmark dataset
[17]	SMOTE + PCA/DWT with SVM	Acc: 97.30%, AUC: 99.62%, F1: 93.08%	Not specified
[18]	GCNN with AE (CAE/VAE)	Acc: 98.89%	NMT-Scalp
[19]	CNN + BiGRU + CBAM	Binary: 99.00%, 3-class: 96.20%, 4-class: 92.00%, 5-class: 89.00%; Sens: 89.00–99.00%, Spec: 89.63–99.00%	Public EEG dataset
[20]	IANFIS-LightGBM	High classification accuracy (specific values not detailed)	University of Bonn, Boston Children's Hospital EEG
[21]	Hybrid AdaBoost, RF, DT with optimization	CHB-MIT: Acc: 96.61%, Sens: 94.67%, Spec: 91.36%; Siena Scalp: Acc: 95.31%, Sens: 93.18%, Spec: 90.07%	CHB-MIT, Siena Scalp
[48]	leveraging Explainable Artificial Intelligence (XAI)	Accuracy: 96.02% , Precision: 95.60%, Recall: 95.87% , F1-score: 95.74%	UBMC dataset
This work	DT-FIS	accuracy of 99.23%, specificity 99.02%, precision of 98.92%, a recall of 99.1%, F1 score of 99.18%	UBMC dataset

6.4. Discussion

A notable increase in accuracy and efficiency is demonstrated by the performance results of the epileptic seizure detection system described in the research employing the DT-FIS hybrid model improved using meta-heuristic algorithms (WCA-AE and SFOA). Using simulation on the UBMC dataset with four classes of EEG signals, this method achieved 99.23% accuracy, 98.92% prediction accuracy, 99.1% sensitivity, and 99.18% F1 score. These results are competitive and better than some other advanced models, including HyEpiSeiD (99.01%) and 1D-CNN with attention mechanism (99.83%), as well as traditional methods. When combined with DT-FIS training by Starfish Optimization Algorithm (SFOA). WCA simulates evaporation and precipitation. It optimizes the AE matrix. This reduces features from 34 to 10.Also, it increase the intra-class correlation. This effectively managed the data's complexity and noise while also increasing accuracy. Following 1800 WCA rounds and 85 SFOA iterations, the error dropped to less than 1% and the clustering's

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node dispersion was optimized, which improved class discrimination, according to the graphs and tables (see Figure 10 and Table 5).

The suggested approach reached its peak in Scenario 6 with an accuracy of 99.23% and an F1 score of 99.28%, according to the performance comparison in various scenarios (Table 5 and Figure 13). This represents a notable improvement over the beginning scenarios (e.g., Scenario 1 with 75.28%). This improvement is especially apparent when the feature reduction matrix is optimized and the DT-FIS system, which combines fuzzy logic and binary decision making, is used. However, if the number of characteristics is drastically reduced (for example, by utilizing only five features), the accuracy drops to 78.28%, highlighting the significance of striking a balance when choosing the number of features. DT-FIS with the UBMC dataset exhibits relative superiority when compared to related works (Table 7). It has great potential for clinical applications, particularly in real-time monitoring and early seizure detection; however, testing on a wider range of populations is still necessary to confirm its generalizability.

Thus, it was discovered that simpler classification techniques can be used to successfully identify epileptic episodes. Furthermore, the explainable feature and channel selection process allow for excellent performance with minimum computational effort. Using this method, we were able to verify without changing the dataset that the selected channels are located in the focal regions of epileptic convulsions. This encourages additional study using custom models, which could further reduce the number of channels and confirm if the focus point is always the best place to collect information for epileptic seizure detection.

This study aims to demonstrate the technical feasibility of this approach to support clinical trials. Further study is needed to include patient demographics and further epileptic episodes in the dataset. The goal is to develop a dynamic compression ratio technique that adapts to patient characteristics and optimizes signal quality in a variety of epileptic populations. Furthermore, the application of state-of-the-art machine learning techniques to enhance the system's adaptability will be examined. By overcoming these limitations and pursuing these research directions, the goal is to improve the method's clinical utility, resilience, and flexibility in epilepsy monitoring and diagnosis, which has the potential to revolutionize the field of portable epilepsy diagnostic tools.

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6.5. Theoretical Justification of DT-FIS with Meta-Heuristic Optimization

The rationale for combining the Decision Tree-based Fuzzy Inference System (DT-FIS) with meta-heuristic optimization algorithms lies in addressing inherent limitations of traditional decision trees when applied to complex, noisy, and nonlinear data such as EEG signals. Decision trees are prone to over fitting due to their greedy splitting mechanism and high sensitivity to small data variations, which can lead to poor generalization in EEG classification tasks characterized by uncertainty and high dimensionality. Integrating fuzzy inference systems mitigates this by introducing soft decision boundaries through Gaussian membership functions, effectively modeling the inherent nonlinearity and uncertainty in EEG signals. This hybrid DT-FIS can be theoretically viewed as a fuzzy entropy-regularized classifier, where fuzzy rules reduce variance and enhance robustness.

To independently quantify the impact of each optimization component, comprehensive ablation studies were conducted under the LOSO cross-validation protocol. Table 15 presents the results, demonstrating the individual and synergistic contributions of WCA-AE (for feature reduction) and SFOA (for parameter training).

Table 15- Ablation Studies on the Impact of Optimization Components

Model Configuration	**Accuracy (%)**	**Improvement over baseline (%)**	**F1-Score (%)**	**Description**
DT-FIS without optimization	85.47	-	84.92	Base DT-FIS with default parameters and full features
DT-FIS + WCA-AE only	93.68	+8.21	92.15	Feature dimensionality

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(feature reduction)				reduction applied, improving intra-class density
DT-FIS + SFOA only (parameter training)	96.32	+10.85	95.78	Fuzzy and tree parameters optimized, reducing over fitting
DT-FIS + WCA-AE + SFOA (full proposed model)	99.12	+13.65	98.95	Synergistic effect of both optimizations

These results highlight that WCA-AE primarily enhances feature quality by increasing intra-class density and inter-class separation, while SFOA focuses on fine-tuning fuzzy rules and tree structure to prevent over fitting. Their combination yields a significant synergistic improvement, validating the proposed hybrid framework both theoretically and empirically.

7. Conclusion and future work

With a 99.23% accuracy and a 99.18% F1 score on the UBMC dataset, the DT-FIS hybrid model with WCA-AE and SFOA meta-heuristic algorithms was presented in this paper for the identification of epileptic seizures. This model outperforms many other approaches currently in use. Through feature dimension reduction and interclass correlation enhancement, this technique greatly increased the CAD system's efficiency in analyzing complicated EEG signals. Results show that this model has a strong potential for early seizure detection and enabling prompt medical interventions, both of which can improve the management of epilepsy. Nevertheless, there are drawbacks, like reliance on a particular dataset and the requirement for validation on a wider range of populations. To improve the model's generalizability, the dataset must be expanded in the future by include demographic information and additional seizure types. Additionally, the quality of diagnosis can be enhanced by creating techniques to dynamically modify the signal compression ratio based on patient variables. The program will also include research into more sophisticated learning algorithms to improve system flexibility and lessen computing load. The ultimate objective is to develop this model into a portable, effective clinical epilepsy diagnosis tool that satisfies patients' actual demands. Innovation in epilepsy monitoring equipment may result from this research.

To distinguish between various states of consciousness, we suggested an EEG analysis technique in this work. To achieve this goal, nurses and family members administered auditory and tactile stimulation to patients in five phases, and EEG signals were monitored throughout each stage. Both pre- and post-stimulus EEG data were recorded. A statistical analysis was performed on the features that were taken out of the EEG signal sub bands for various GCSs. This study demonstrates that coma patients with GCS 3 and GCS 8 differ from other states of consciousness in terms of lower sub theta energy values, even though the features were successful in differentiating the majority of GCSs. According to system training using the Harris Hawks optimization method, the suggested classification, which is based on three combined machine learning techniques—Fuzzy-PCA, FSVM, and Fuzzy K-NN—has attained excellent accuracy. The suggested recording and analysis strategy has allowed this study to categorize patients in deep coma (GCS between 3 and 8) with a performance of 99.53% and 98.7% classification accuracy. Based on the findings that EEG signals may successfully distinguish between LeOC and profound coma, this study is applicable with a very lightweight and high-accuracy classification system method.

Declarations

Ethical Approval Not applicable.

Competing interests The authors have no competing interests, or other interests that might be perceived to influence the results and/or discussion reported in this paper.

Authors' contributions All authors contributed to the study conception and design. Data collection, simulation and analysis were performed by Mahdiyeh lak, Jasem Jamali, Mehdi Taghizadeh, Nahid Adlband

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and Omid Mahdiyar. The first draft of the manuscript was written by Jasem Jamali and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Availability of data : The data used in the paper will be available upon request. Please contact jasemjamali54@gmail.com.

List of Abbreviations

Abbreviation	Full Form	Description (if needed)
EEG	Electroencephalogram	Brain electrical activity recording
DT-FIS	Decision Tree-based Fuzzy Inference System	Proposed hybrid classifier
SFOA	Starfish Optimization Algorithm	Meta-heuristic for model parameter optimization
WCA	Water Cycle Algorithm	Meta-heuristic for feature reduction
WCA-AE	Water Cycle Algorithm-optimized Auto-Encoder	Feature mapping and dimensionality reduction
CAD	Computer-Aided Diagnosis	Automated diagnostic system
LOSO	Leave-One-Subject-Out	Patient-independent cross-validation
GCS	Glasgow Coma Scale	Clinical scale for assessing consciousness
AUBMC	American University of Beirut Medical Center	Source of the utilized EEG dataset
WHO	World Health Organization	–
CNN	Convolutional Neural Network	–
GRU	Gated Recurrent Unit	–
HHO	Harris Hawks Optimization	Alternative meta-heuristic used in coma part
FSVM	Fuzzy Support Vector Machine	–
PCA	Principal Component Analysis	–
SNR	Signal-to-Noise Ratio	–
SD	Standard Deviation	–

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